YAKEEN

NEET 2.0 2027

Physics | Lecture - 10

Motion in a Straight Line

By- Manish Raj (MR Sir)



Todays Goal

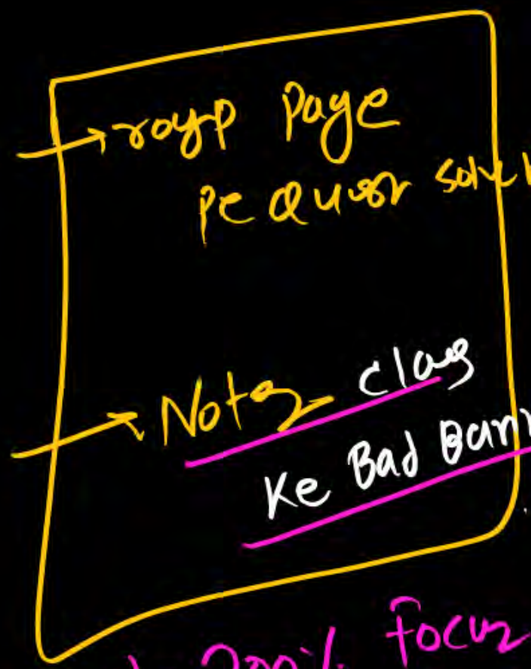
PhD on graph → 2

HW goal

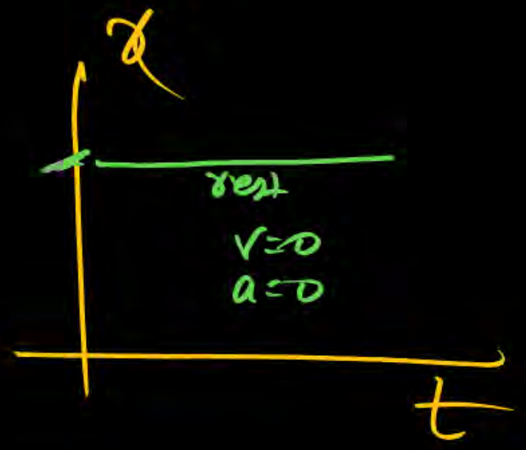
→ 1-D PYQ given in this Noty

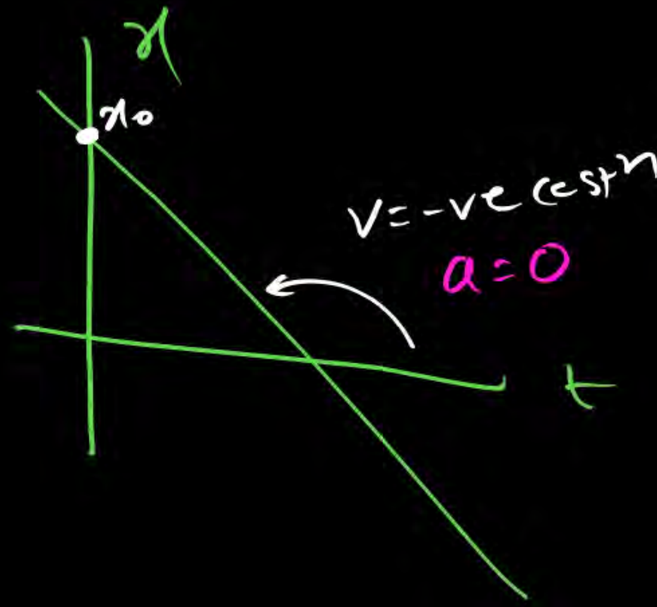
→ MMQ → last Page of PPT

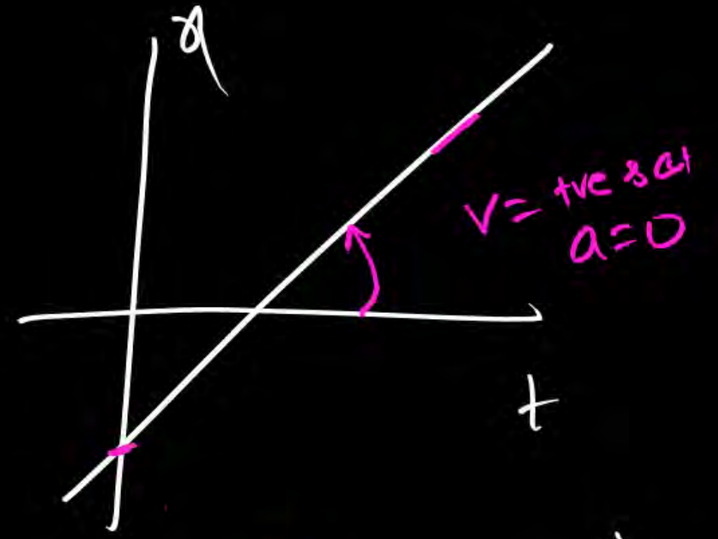
→ Assignment → 4 of 1-D

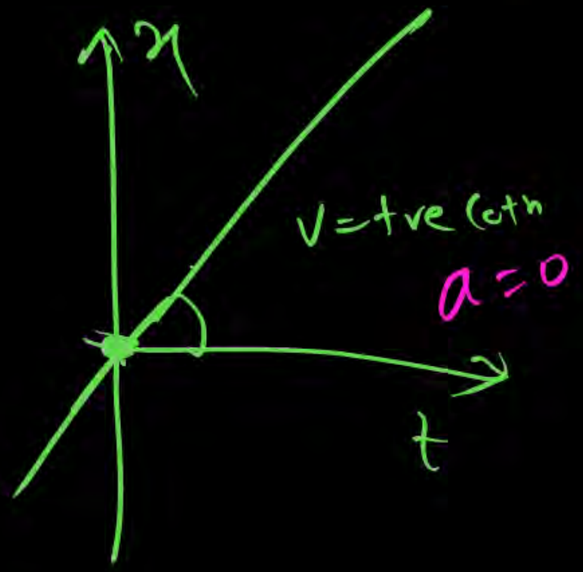


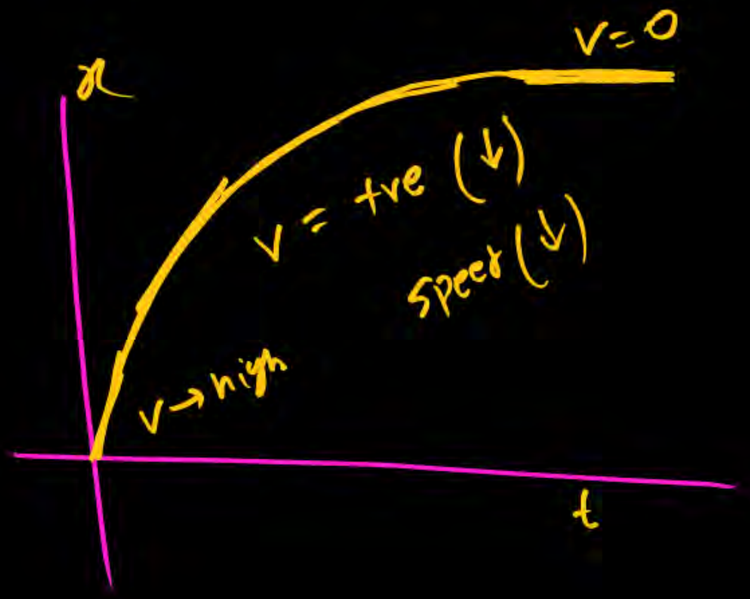
200% focus for todays class

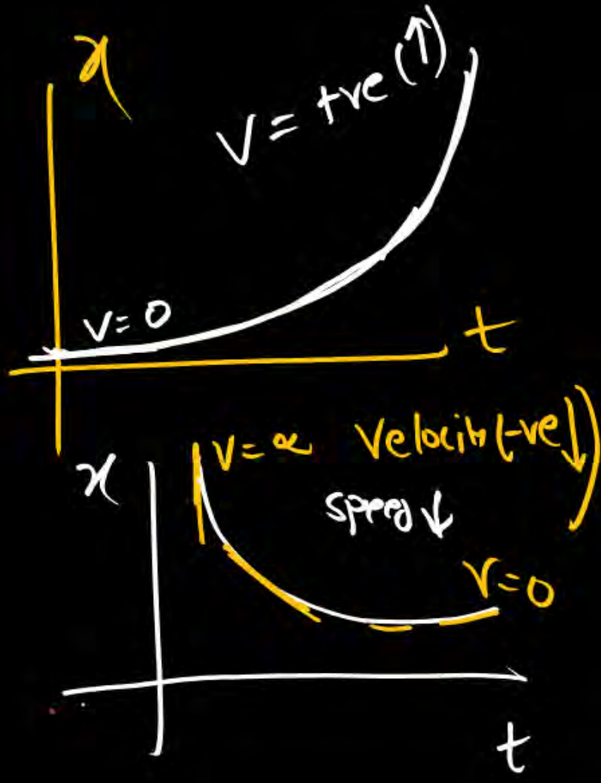

x
rest
v=0
a=0
t


x
x0
v=-ve (constn)
a=0
t

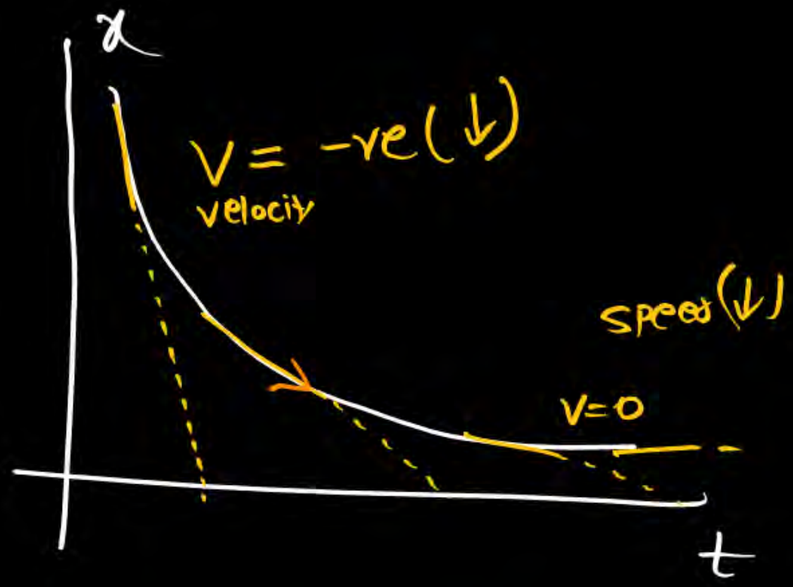

x
v=+ve & ↑
a=0
t


x
v=+ve (const)
a=0
t


x
v=0
v = +ve (↓)
speed (↓)
v→high
t


x
v = +ve (↑)
v=0
t
x
v=∞ Velocity (-ve↓)
speed↓
v=0
t

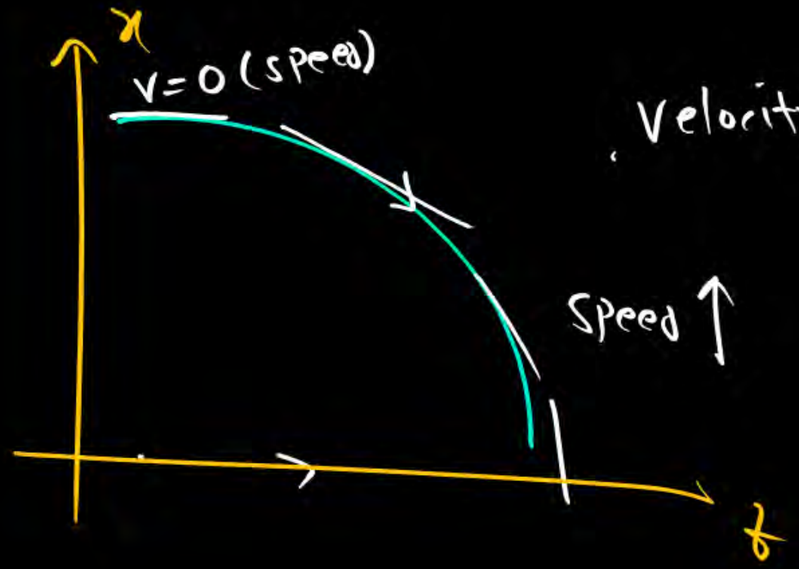
Cop 1



acc^{n} opposite to velocity a = +ve

retarding motn → slow down

#



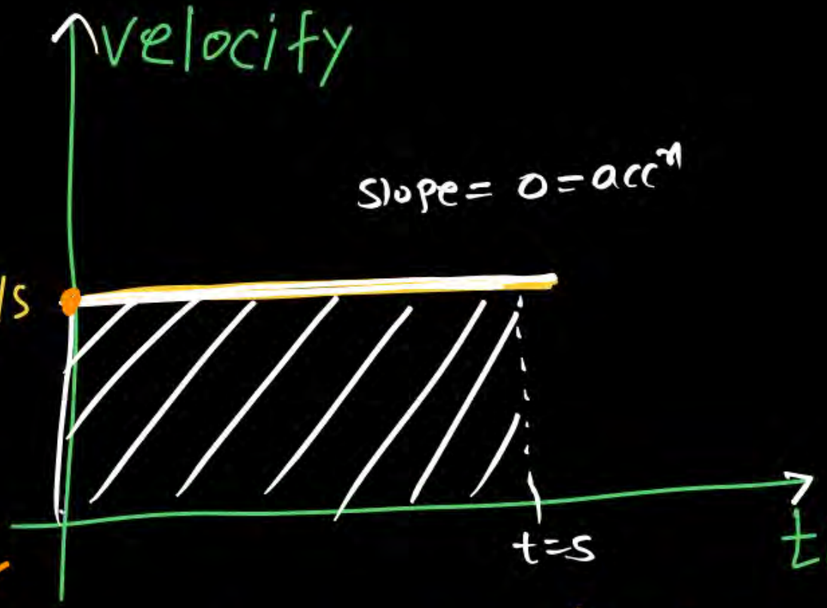
Velocity = -ve (↑)

direction

speed ↑ (acc^{n} ||^{y} to velocity = -ve)

Velocify time Graph :-

(15)



dist^n = disp^m

V = cost^n (Uniform motion)

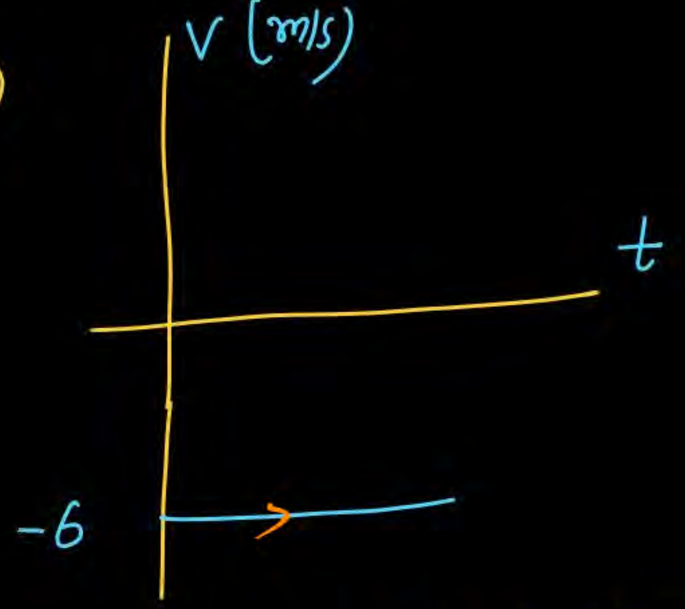
(No-U-turn)

a = 0

find disp^m in 5-sec ??

= Area of v/t graph = 10 x 5 = 50m ✓

(16)



V = -6m/s (cost^n velocity)

→ uniform motn

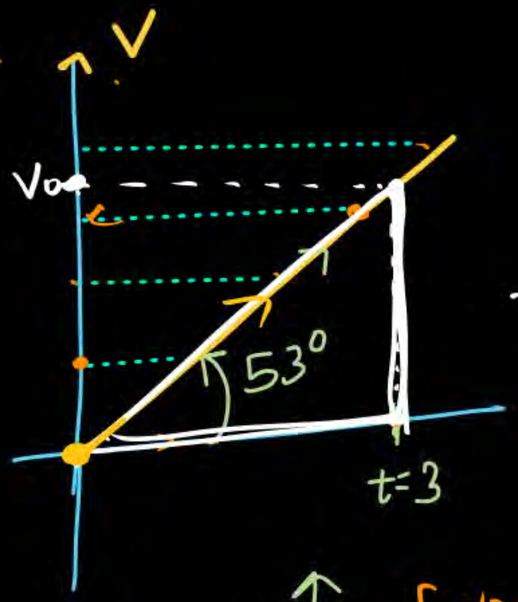
→ a = 0

disp^m in 3-sec

S = u x t = -3 x 6 = -18m

↑ dirn of motn

(17)



$\tan 53^\circ = \frac{4}{3} = \frac{V_0}{3}$

$V_0 = 4$

Area $= \frac{1}{2} \times 3(4) = 6m$

velocity $= +V \uparrow$ [No-U-turn]

$a = \tan 53^\circ$ (slope) $= +\frac{4}{3}$ (const)

disp^m in 3-sec, $u_i (t=0) = 0$

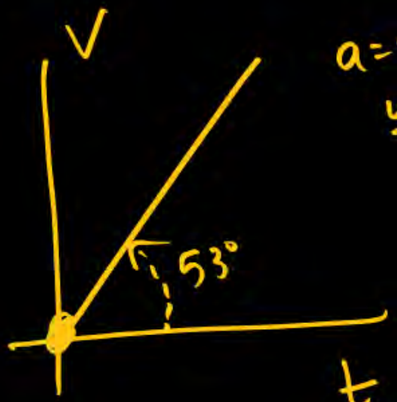
$a = \frac{4}{3}$

$\hookrightarrow S = ut + \frac{1}{2}at^2$

$= 0 + \frac{1}{2}\frac{4}{3}(3)^2 = \frac{2}{3} \times 9^3 = 6m$

motion starts from rest and const^n acc^n $a = \frac{4}{3} m/s^2$ (const)

draw V/t graph



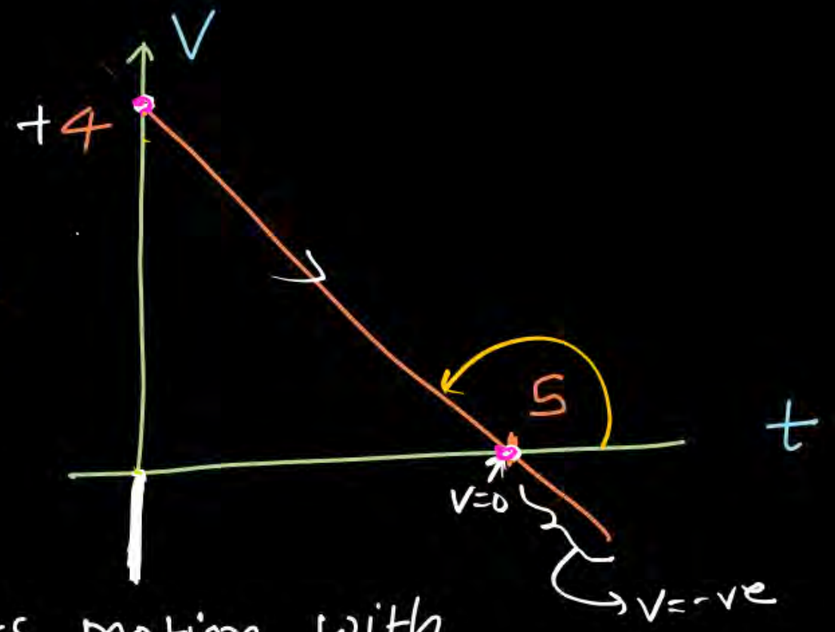
a = slope = tanθ

$\frac{4}{3} = \tan\theta$

(18)

decode this graph

find Sab Kuchh



Soln

object starts motion with intial velocity 4 m/s and acc^n $-4/5\ m/s^2$ (Cost^n)

$\Rightarrow$ Velocity (+ve upto 5 sec) $\downarrow$

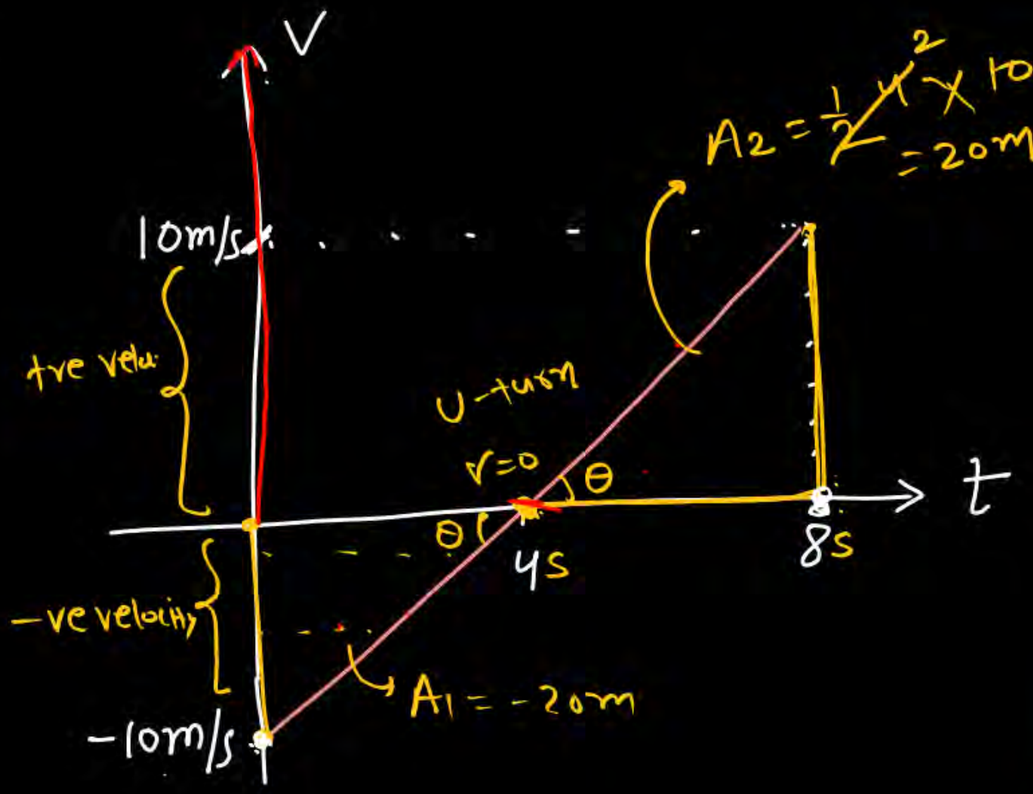
$\Rightarrow$ Velocity is zero at t = 5 sec

acc^n = Slope = $-\frac{4}{5}\ m/s^2$

disp^m in 5-sec = $\frac{1}{2} \times \frac{4}{5} \times 5$

also dist^n = +10m

19.



Soln

object starts his motion with $-10\,m/s$ and (constant accn $= \frac{10}{4} = \frac{5}{2}\,m/s^2$

$a = ?? = \frac{5}{2}\,m/s^2$

disp^n in 8-sec $= 0$

Avg velocity in 8 sec $= 0$

(#) Avg speed in 8-sec $= \frac{\text{total disp}^n}{\text{total time}}$

(#) Avg accn in t = 8sec $= \frac{\vec{V}_f - \vec{V}_i}{\Delta t} = \frac{10-(-10)}{8} = \frac{20}{8} = \frac{5}{2}\,m/s^2$

$= \frac{40}{8} = 5\,m/sec.$

(20) disp^m of particle is maximum at

(a) t_1

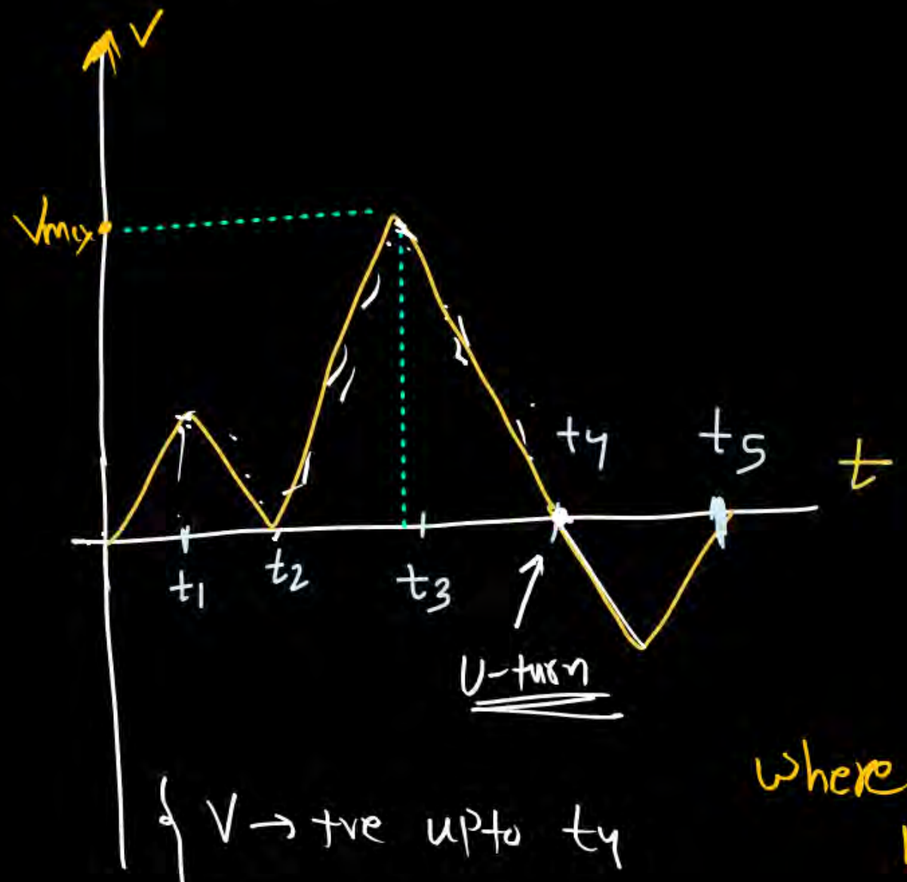
(b) t_2

(c) t_3

(d) ✓ t_4

(e) t_5

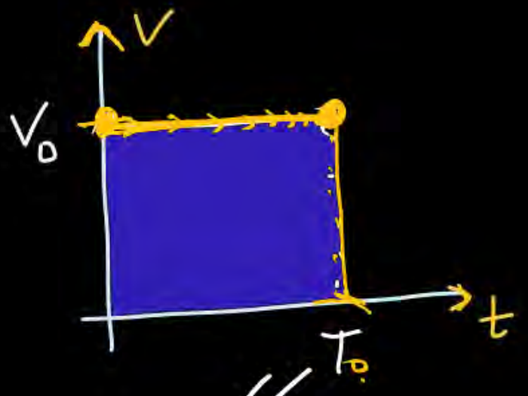
$\text{disp}^m_{max} = \text{Area}_{max}$ → at t_4



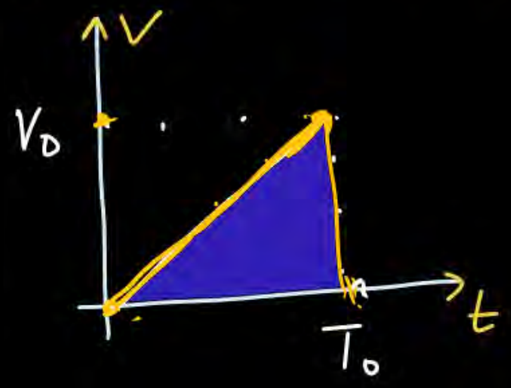
{ V → +ve upto t_4
V → -ve aftr t_4

where velocity is max^m → at t_3 ✓

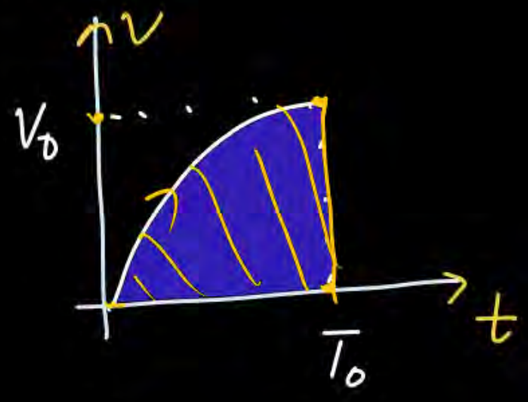
(21) Avg. velocity is maximum in which graph:-



(a) 36%



(b)



(c)

(d) same in all.

33%

$$\text{Avg velocity}_{max} = \frac{\text{Total displ}^m}{\text{total time}} = \frac{(\text{Area})_{max}}{(T_0)_{same}}$$

Area of v/t graph = displ^m

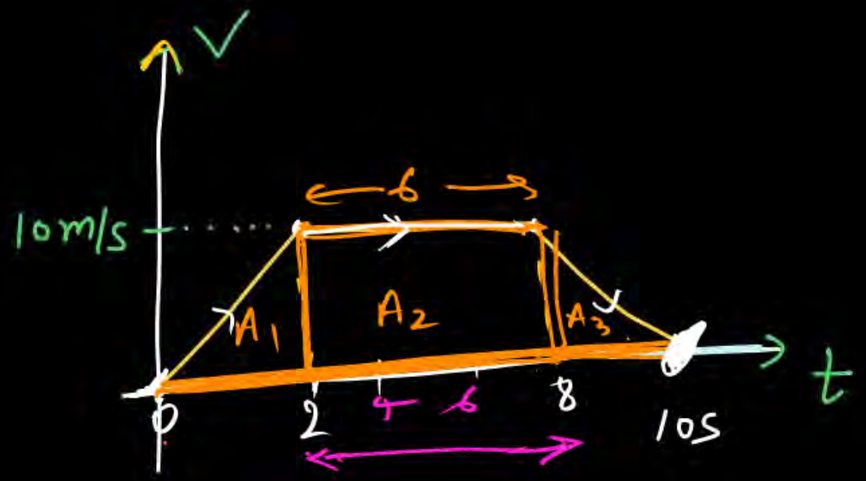
No pen

$$\text{avg acc}^n = \frac{\vec{v}_f - \vec{v}_i}{\Delta t}$$

(i) $a_{Avg} = 0$ in graph a)

(ii) a_{Avg} = same in b & c $= \frac{V_0 - 0}{T_0 - 0} = \frac{V_0}{T_0}$

(22)



Soln

Total displ in 1osec $= \frac{1}{2}(10+6) \times 5$

$= 80m$ ✓

$A_1 = \frac{1}{2} \times 2 \times 10 = 10m$

$A_2 = 6 \times 10 = 60m$

$A_3 = \frac{1}{2} \times 2 \times 10 = 10m$

} moved

$a\,(t=0 \text{ to } 2sec) = +ve = \frac{10}{2} = +5m/s^2$ ✓

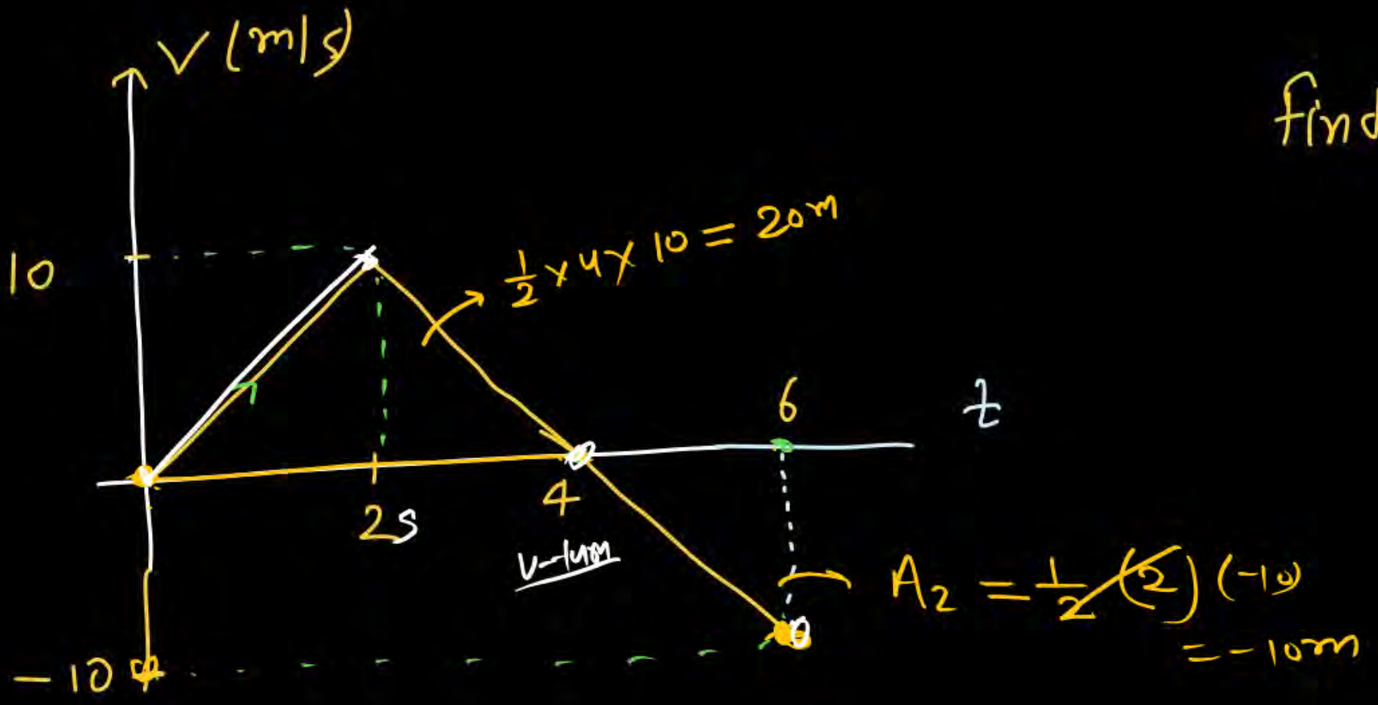
$a\,(2 \text{ to } 8sec) = 0$ uniform motn

$a\,(8 \text{ to } 10s) = -ve$ ✓

$\vec{a}_{\text{Avg acc in 1osec}} = \frac{\vec{v}_f - \vec{v}_i}{\Delta t} = \frac{0-0}{10}$

$= 0 m/s^2$

(23.)



find disp^m in 6-sec

disp^m = Area of v/t graph = 20 - 10 = 10m

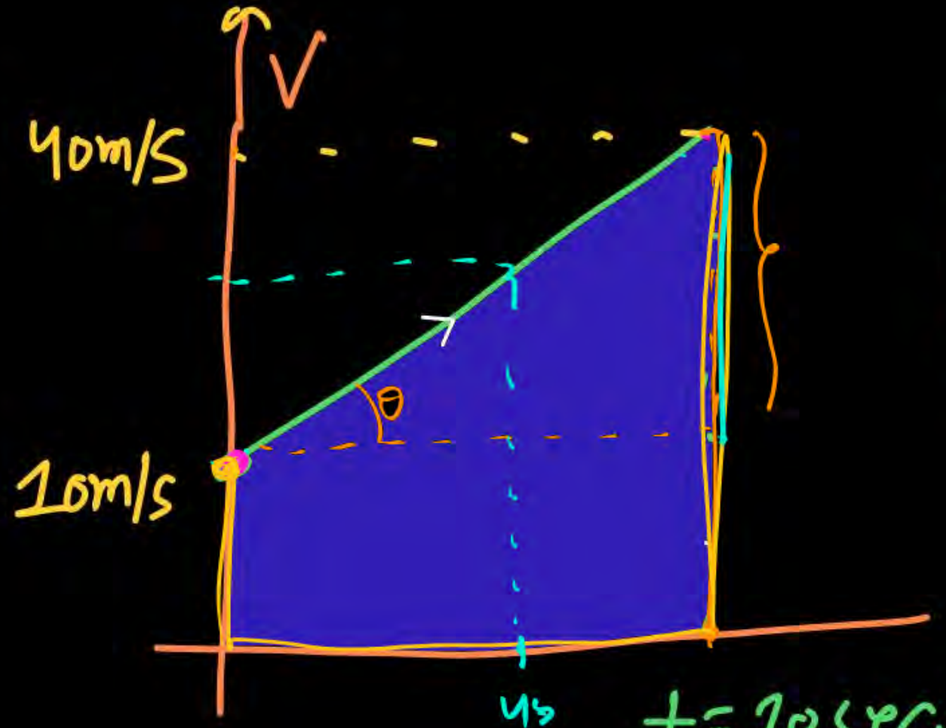
disp^n = 20 + 10 = 30m

Avg velocity in 6 sec = $\frac{10}{6}$

Avg speed = $\frac{30}{6}$

Avg acc^n in 6sec = $\frac{-10-0}{6} = \frac{-10}{6}$

(24)



$y = mx + c$

$V = 3t + 10$ — (1)

$V = 3 \times 4 + 10$

$V = 22 m/s$

velocity at 4sec

$v = u + at$ ✓

find acⁿ & velocity at $t = 10sec$ & displacement in 10sec. ✓

$V_{t=10sec} = 40 m/s$

$a_{const} = +ve\ coef = \frac{40-10}{10}$

$= 3 m/s^2$

⇒ Motion Starts from rest ~~X~~

" " with 10m/s ✓

NO U-turn ✓

$S = 10 \times 10 + \frac{1}{2} 3 (10)^2$

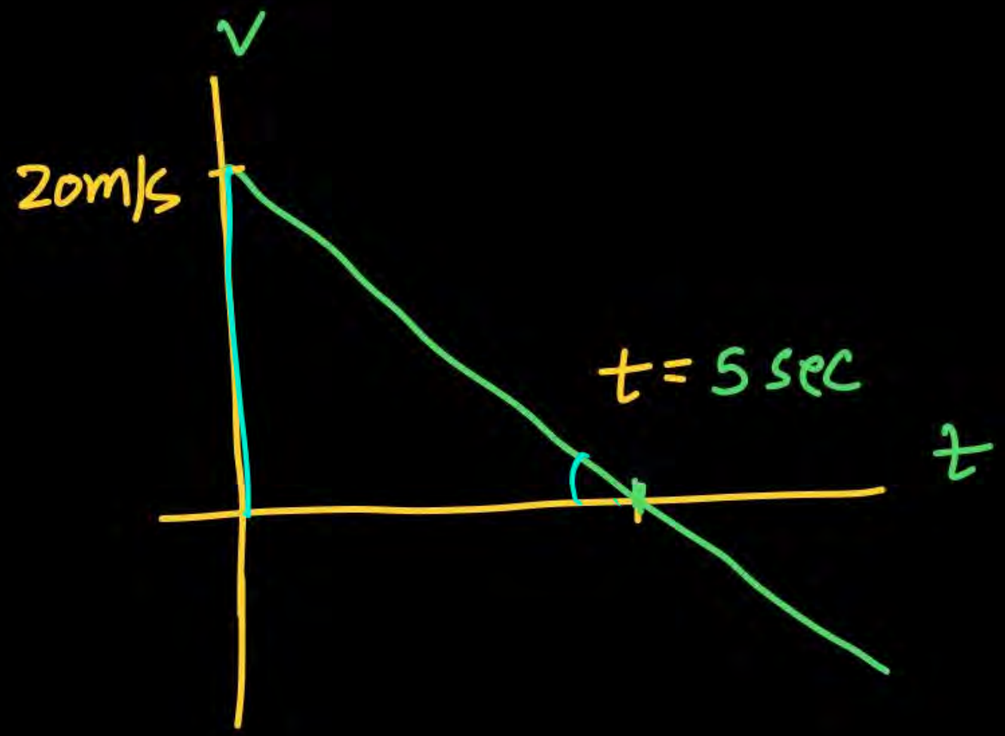
$= 100 + \frac{3}{2} \times 100^{50}$

$= 100 + 150 = 250m$

$Area = \frac{1}{2} (40+10) \times 10^{5}$

$= 50 \times 5 = 250m$ ✓

(25)



(a) find velocity at t = 4sec

a = -ve coeff

$= -\frac{20}{5} = -4 m/s^2$

u = 20m/s

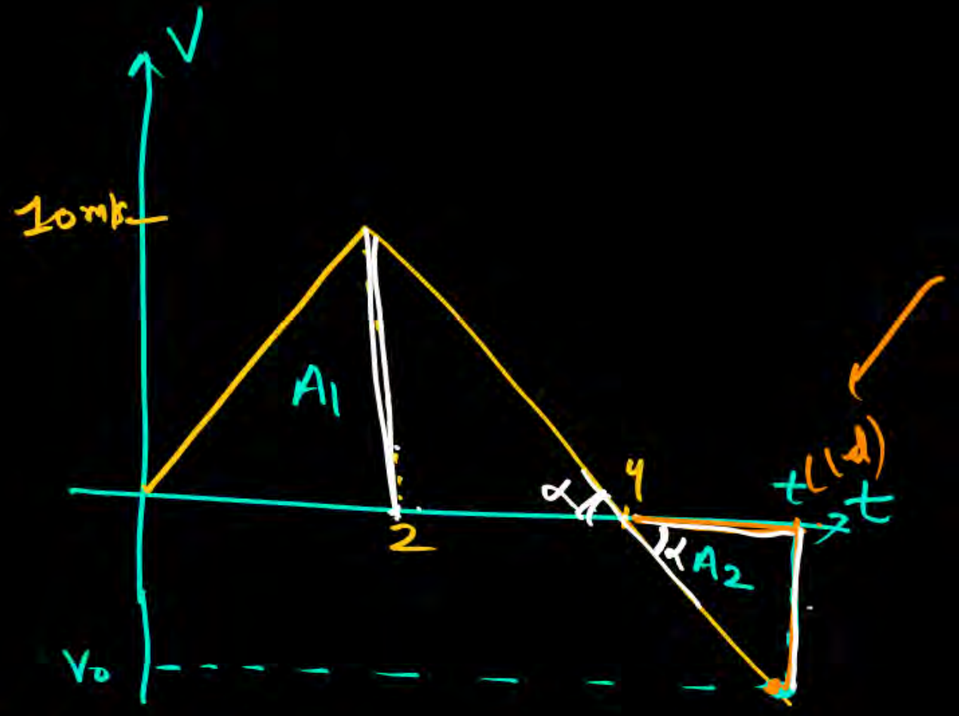
$v = u + at$ $y = c + mx$

$V = 20 - 4t$ ✓

$V = 20 - 4 \times 4$

$= 20 - 16 = 4 m/s$

(26)



(a) find time at which Avg. velocity of Particle is zero.

$$\text{Avg velocity} = \frac{\text{Total displ}^n}{\text{total time}} = 0$$

$$= A_1 + A_2 = 0$$

$$A_1 = -A_2$$

$$A_1 = \frac{1}{2} \times 4 \times 10 = 20\text{m} \quad (+\text{ve Area})$$

$$A_2 = \frac{1}{2}(t-4)V_0$$

$$40 = (t-4)V_0 \quad \text{put } V$$

$$8 \times 10 = (t-4)\,5(t-4)$$

$$8 = (t-4)^2$$

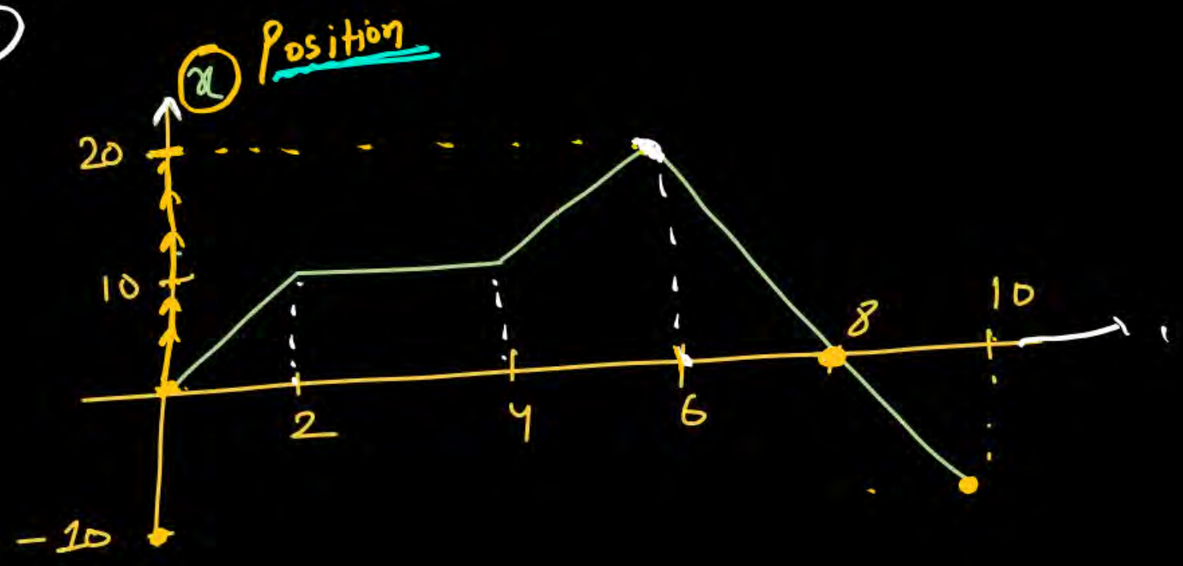
$$t-4 = \sqrt{8} = 2\sqrt{2}$$

$$t = (4+2\sqrt{2})\text{s}$$

$$\tan\alpha = \frac{10}{2} = \frac{V_0}{t-4}$$

$$V_0 = 5(t-4)$$

(27)



Sol'

(i) Avg Velocity in 6-sec $= \frac{\text{total displ}^n}{\text{total tim.}} = \frac{20}{6}$

" " " 10-sec $= \frac{-10-0}{10} = \frac{-10}{10}$

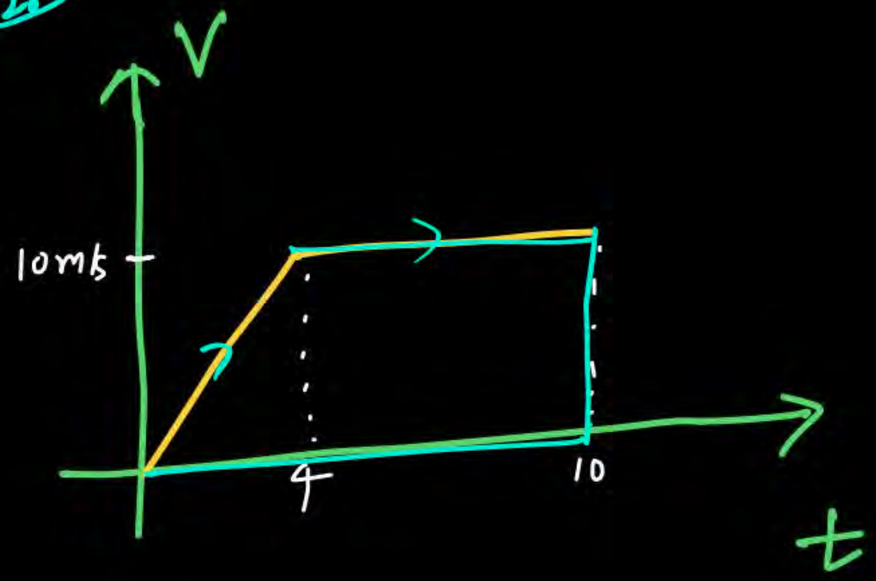
" " " 8 sec $= 0$

Avg speed in 6-sec

" " " 10-sec

" " " 8 sec

(12)

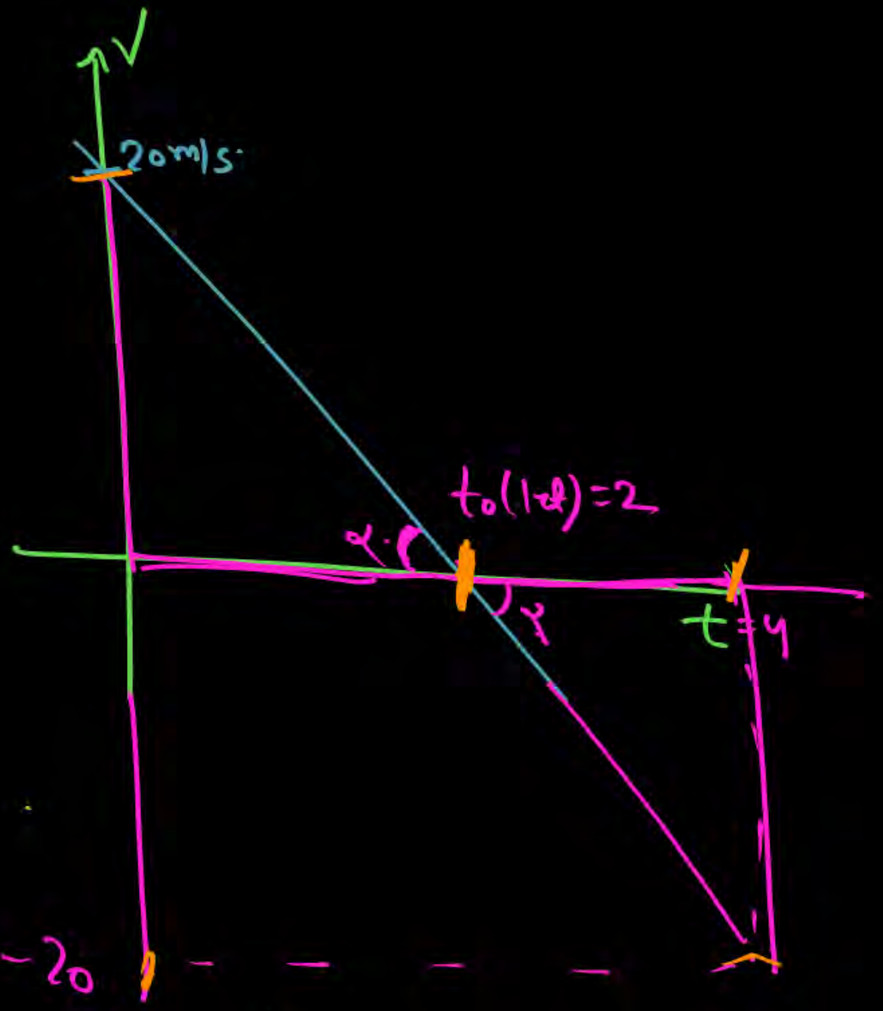


$$\text{Area} = \frac{1}{2} \times (10+6) \times 10$$

Application of graph

(29) Object starts motion with velocity 20m/s & retardation $-10 m/s^2$ find distance & displacement in 4 sec.

Soln



$u = 20 m/s$

$a = -10 m/s^2$ (slope of v/t graph -ve)

$v = u + at$

$0 = 20 - 10t$

$t = 2 sec$ ✓

$v = u + at = 20 - 10 \times 4$

$= 20 - 40 = -20 m/s$

(30) Object starts his motion from rest and constant acceleration $8 m/s^2$ then find displacement in 5 sec

Sol^n

m-1

$u = 0$

$a = 8 m/s^2$

$t = 5 sec$

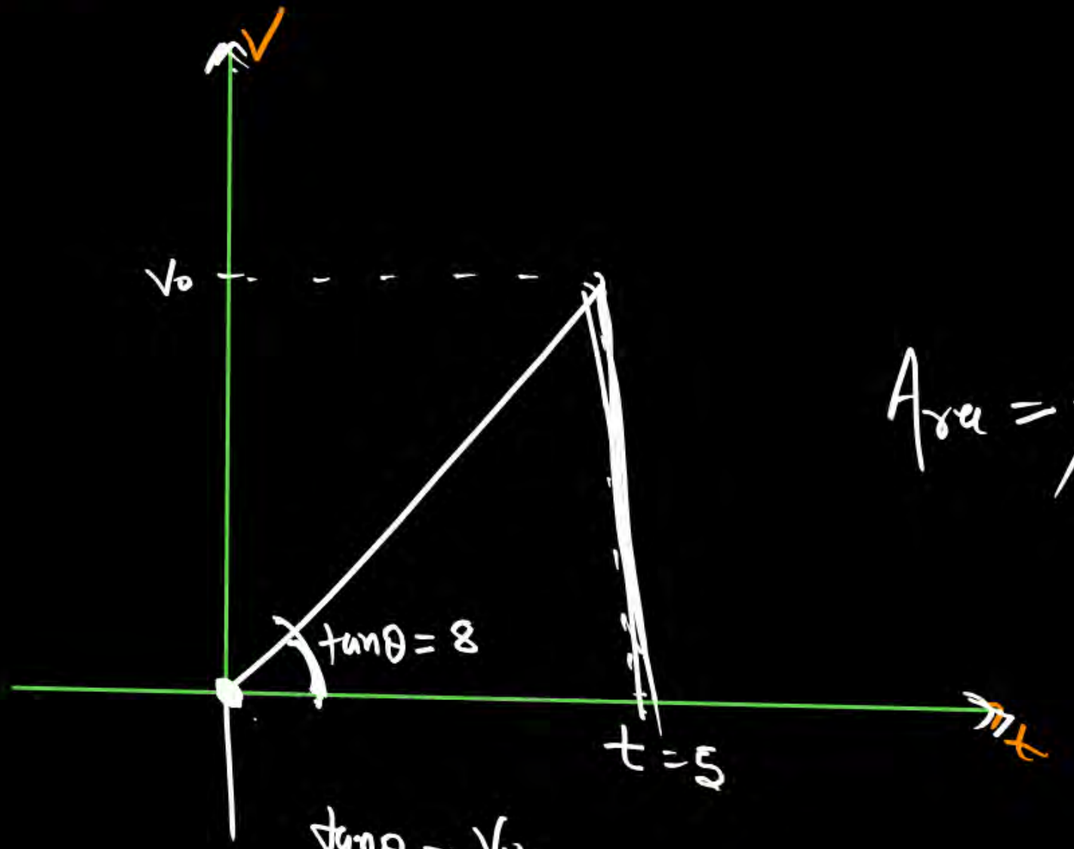
$S = ??$

$S = ut + \frac{1}{2}at^2$

$= 0 + \frac{1}{2}(8)(5)^2$

$= 100 m$

m-2



$\tan\theta = \frac{v_0}{5}$

$8 = \frac{v_0}{5}$

$v_0 = 40$

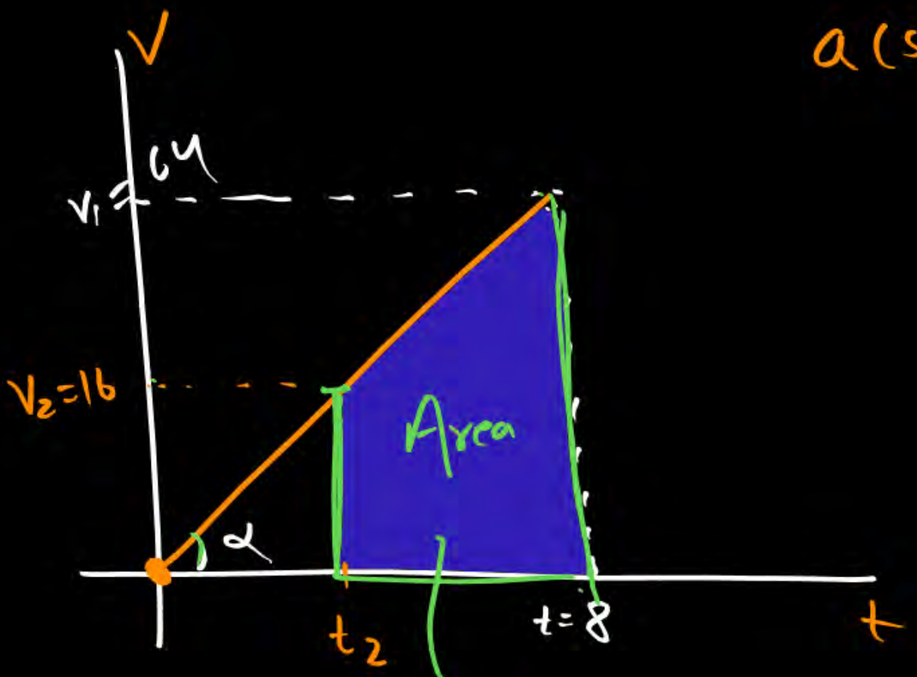
$Area = \frac{1}{2} \times 5 \times 40$

$= 100 m$

3

Object starts his motion from rest and constant acceleration $8\ m/s^2$ then find displacement b/w $t=2$ sec to $t=8$ sec

Soln



a (slope) $= 8 = \frac{v_1}{8}$; $v_1 = 64\ m/s$

$8 = \frac{v_2}{2}$

$v_2 = 16$ ✓

Area $= \frac{1}{2} \times (64+16) \times 6^3$

$= 80 \times 3 = 240\ m$

Q 32

A Particle starts from rest and moves with constant acceleration $4 m/s^2$ for 10 sec after that it moves with constant velocity for 20 more sec, then it retards with $8 m/s^2$ and stop. find total distance moved by object.

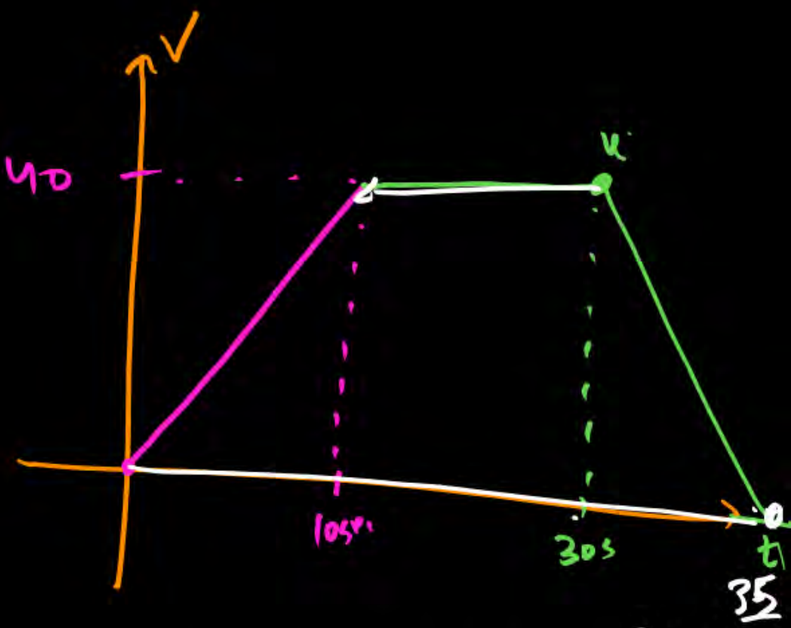
Soln

$u_i = 0$ (rest)
$a = +4 m/s^2$
$t = 10 sec$

$V = u + at$
$= 0 + 4 \times 10$
$V_{10} = 40 m/s$

$V = V_{max}$ constant
$t = 20 sec$
$a = 0$

$u_i = +40 m/s$
$a = -8 m/s^2$
$t = ??$
$V_f = rest$
$t = 5 sec$



$A_M = \frac{1}{2} \times (35+70) \times 40$ (20)

$= 55 \times 20$

$= 1100 m$

33. Initial velocity of object is 60 m/s and retardation $-10 m/s^2$ find distance moved by object is 8-sec.

Soln

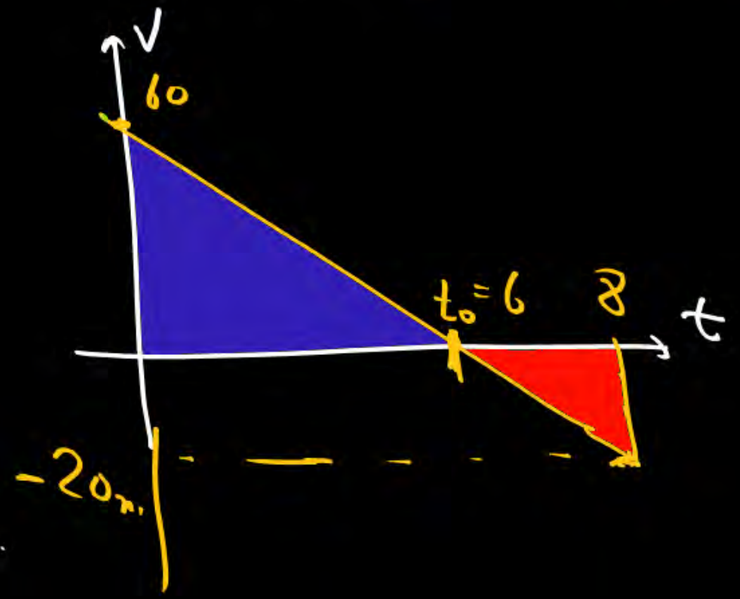
$u = +60 m/s$

$a = -10 m/s^2$

$V = u + at$

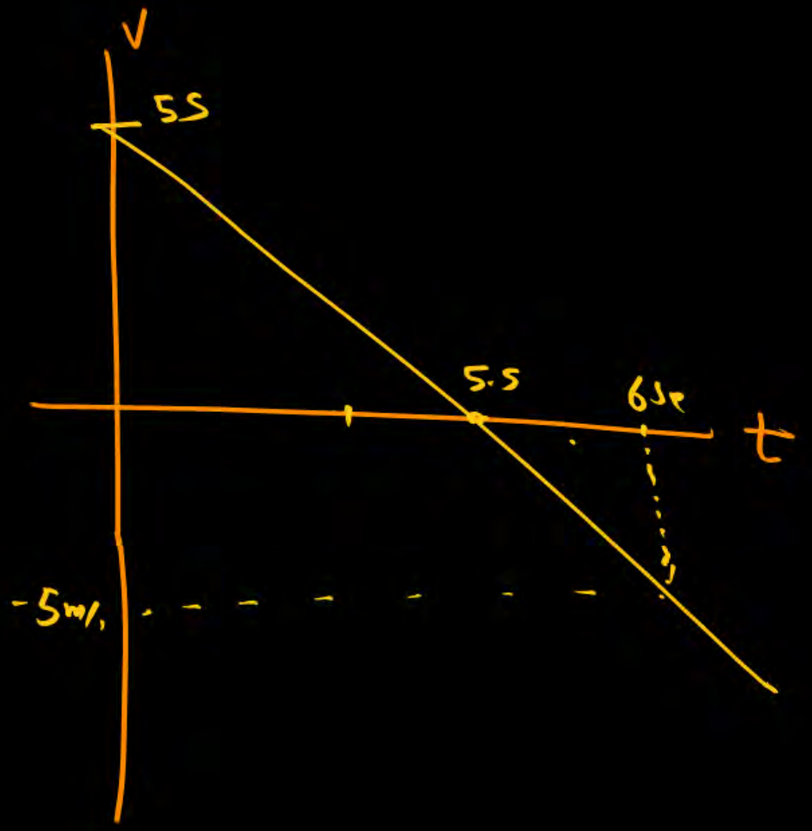
$0 = 60 - 10t$

$t = 6 sec.$



34

CAR is moving with intial speed 55 m/s and retardation -10 m/s^2 find distance in 6th sec.

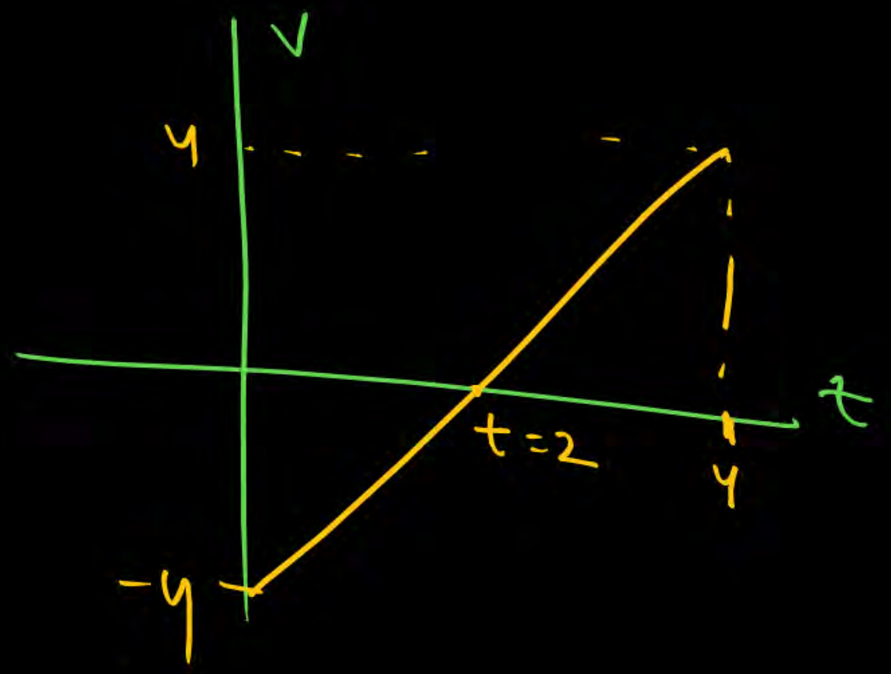


Q 35

Position of object $x = t^2 - 4t + 5$ find distance and displacement in 4-sec.

Soln

$v = 2t - 4$



Question

36

Object starts his motion from rest and constant acceleration α for some time t_1; it retards with β and comes to at rest in time t_2 then find total distance travelled and V_{max} = ?

[IMPORTANT]

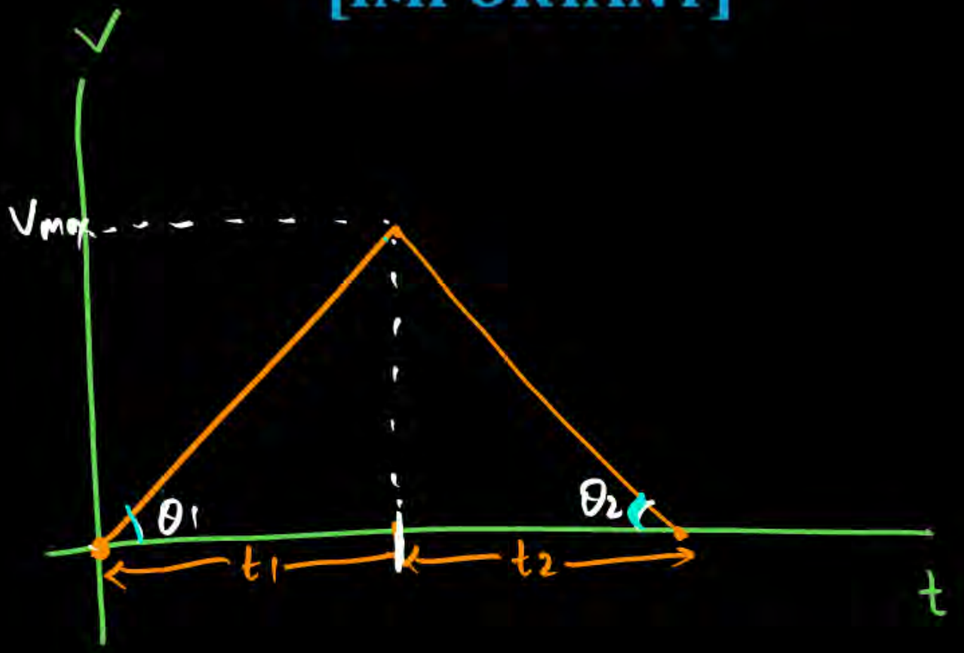
$u = 0$

$a_{(+)} = \alpha$

$t = t_1$

$a_{(+)} = -\beta$

t_2



$\tan\theta_1 = \frac{V_{max}}{t_1}$

$\alpha = \frac{V_{max}}{t_1}$

$V_{max} = \alpha t_1$ - (i)

$\tan\theta_2 = \frac{V_{max}}{t_2}$

$V_{max} = \beta t_2$ - (ii)

$\alpha t_1 = \beta t_2$

$\Rightarrow S = \frac{1}{2} \times (t_1 + t_2) V_{max}$ ✓

$t_1 + t_2 = T$ (Total time of m+n)

$\frac{V_{max}}{\alpha} + \frac{V_{max}}{\beta} = T$ (Put t_1 & t_2 from eq ① & ②)

$V_{max}\left(\frac{1}{\alpha} + \frac{1}{\beta}\right) = T$

$$V_{max} = \left(\frac{\alpha\beta}{\alpha+\beta}\right) T \quad *$$

Put V_{max} in disp'

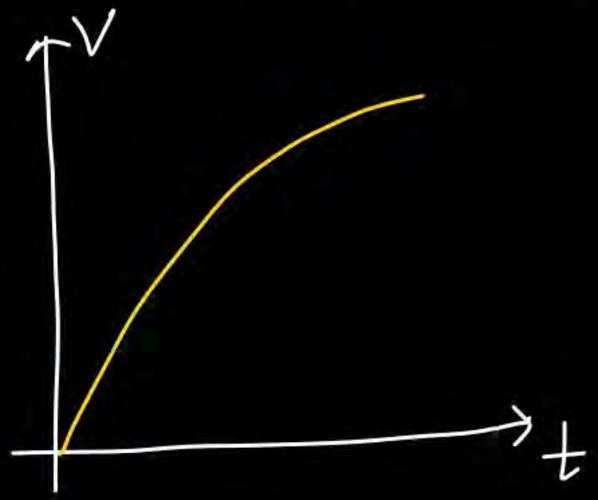
$$S = \frac{1}{2}(t_1+t_2)\, V_{max}$$

$$= \frac{1}{2} T \left(\frac{\alpha\beta}{\alpha+\beta}\right) T$$

$$S = \frac{1}{2}\left(\frac{\alpha\beta}{\alpha+\beta}\right) T^2$$

Yaha pure me ek Barabar Rahega

for given velocity-time graph which of the following acc-time graph is correct



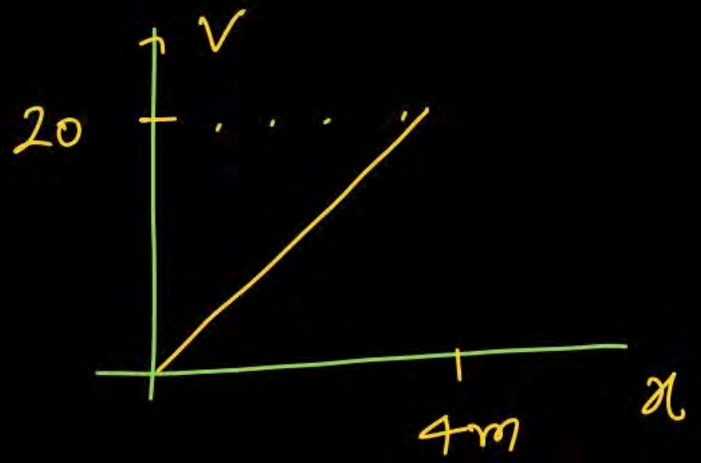




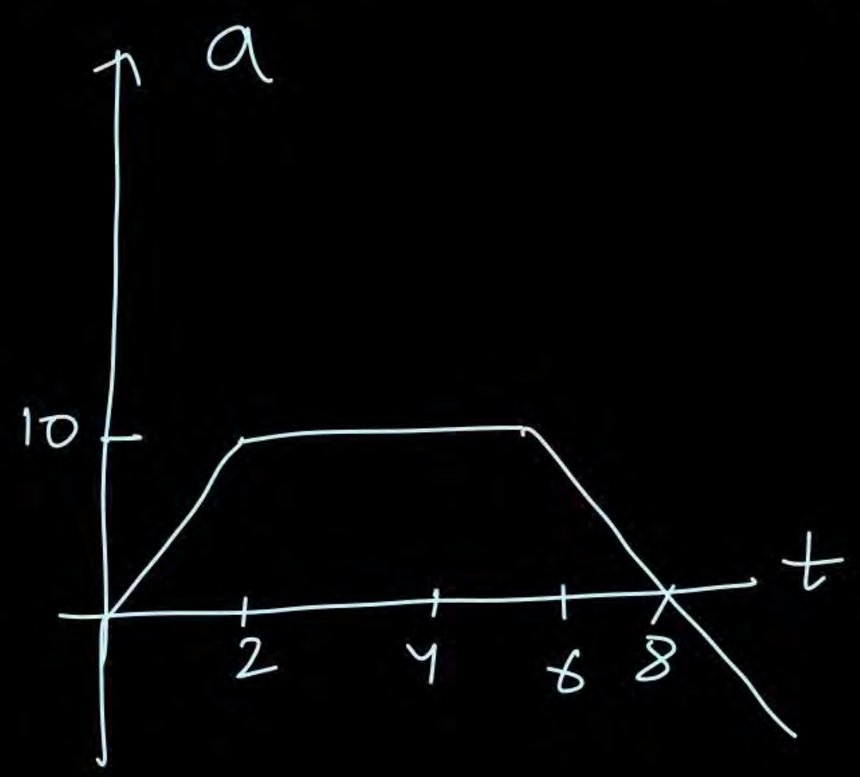




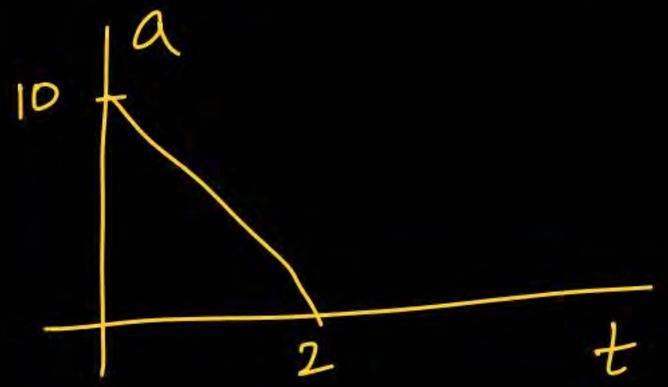
find accn at $x = 1m$



9f intial Velocity of object – 10m/s then find velocity at t = 8sec



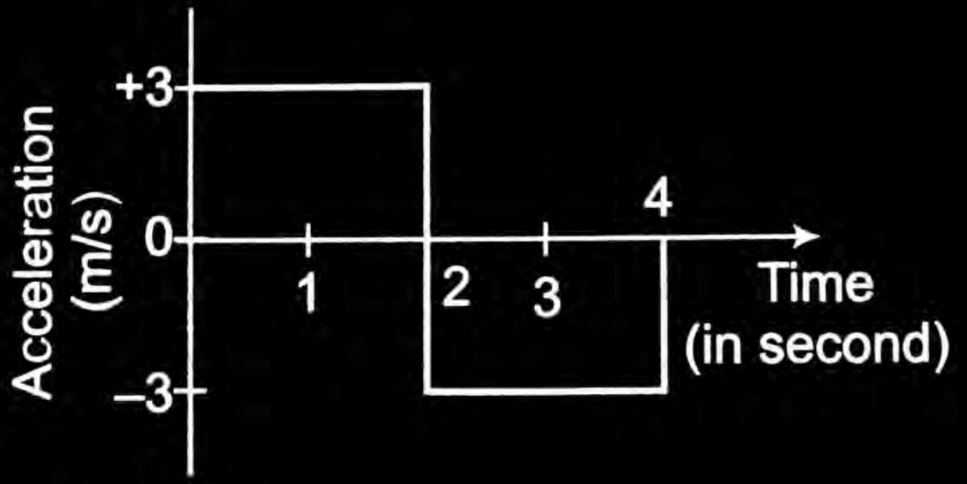
acceleration - time graph:-



if intial velocity is 2m/s then find velocity at $t = 2sec$

Question

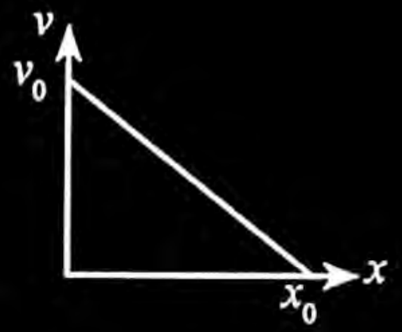
A particle starts from rest at $t = 0$ and moves in a straight line with an acceleration as shown below. The velocity of the particle at $t = 3$s is



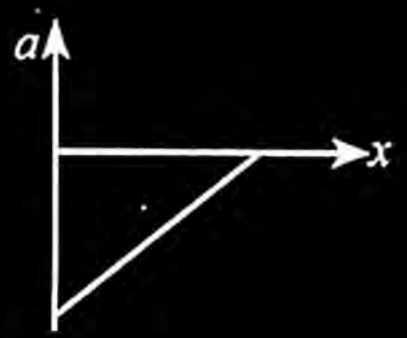
1. 2 m/s
2. 3 m/s
3. 4 m/s
4. 6 m/s

Question

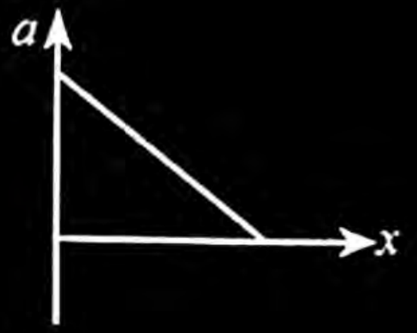
The given graph shows the variation of velocity with displacement. Which one of the graph given below correctly represents the variation of acceleration with displacement? **[JEE Adv. 2015]**



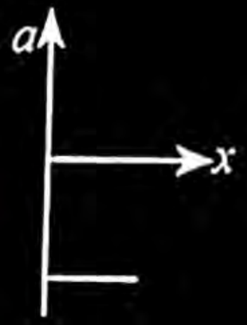
1



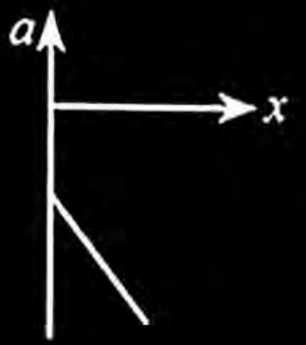
2



3



4



all above question

copy in note

Solve in rough Page

HW

A NO car starts from rest and constant acc^{n} = 10 m/s^2 for 5sec after this it retards with −5 m/s^2 comes to at rest find total displacement.

Question

A particle starts from rest, accelerates at 2 m/s^2 for 10s and then goes with constant speed for 30s and then decelerates at 4 m/s^2 till it stops. What is the distance travelled by it

1. 750 m
2. 800 m
3. 700 m
4. 850 m

NO COPY

Object moving with 20m/s const^{n} velocity for 5-sec after that it retards with $-10m/s^2$ for 4-sec then moves with constant velocity for 2-more sec find distn & disp^{m}

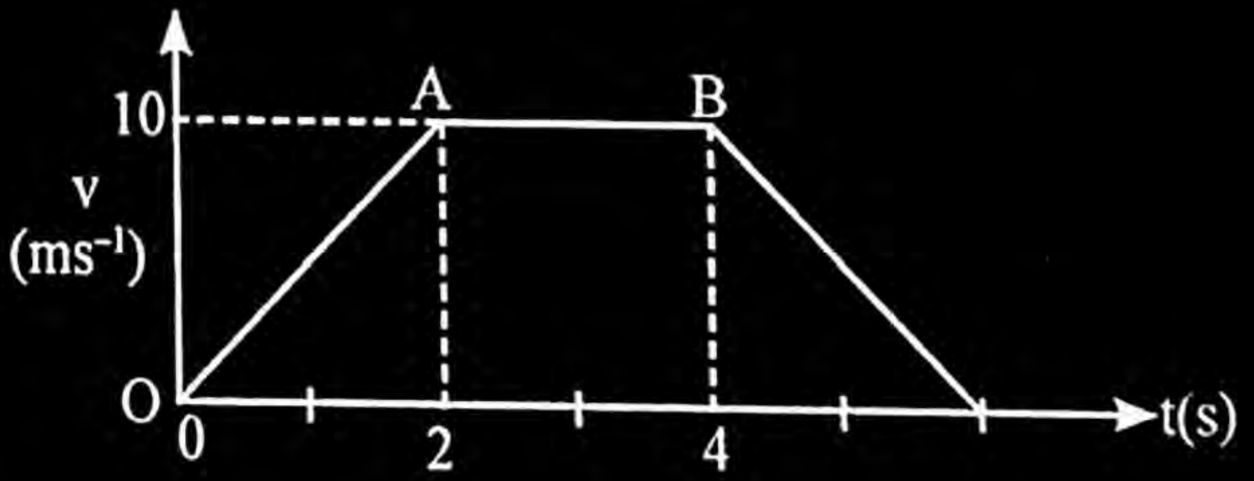
$a = \left(\frac{dv}{dx}\right) v$

	Graph	slope	Area	y-component	x-component
1.	$x - t$ Position – time	$v = \frac{dx}{dt}$	Area $= \int x\,dt$	Position	time
2.	velocity - time $v - t$	$a = \frac{dv}{dt}$	$= \int v\,dt$	velocity	time
3.	velocity – Position $v - x$	Slope $= \frac{dv}{dx} = \frac{a}{v}$	$= \int v\,dx$	veloci	posit.
4.	$v^2 - x$	Slope $= \frac{dv^2}{dx} = 2v\frac{dv}{dx}$ Slope $= 2 \times a$	$\int v^2\,dx$		
5.	acc^n – time $a - t$	Slop $= \frac{da}{dt} =$ [illegible]	$\int a\,dt$	acc^n	tim
6.	acc^n Position $a - x$	[illegible] $= \frac{da}{dx}$	$\int a\,dx$	acc^n	positio.

Graph	slope	Area	y-component	x-component
x - t Position - time			Position	time
velocity - time v - t				
velocity - position v - x				
$v^2 - x$				
acc^n - time a - t				
acc^n position a - x				

Question

The velocity-time graph of an object moving along a straight line is shown in figure. What is the distance covered by the object between $t = 0$ to $t = 4$s? **[JEE Main 2025]**

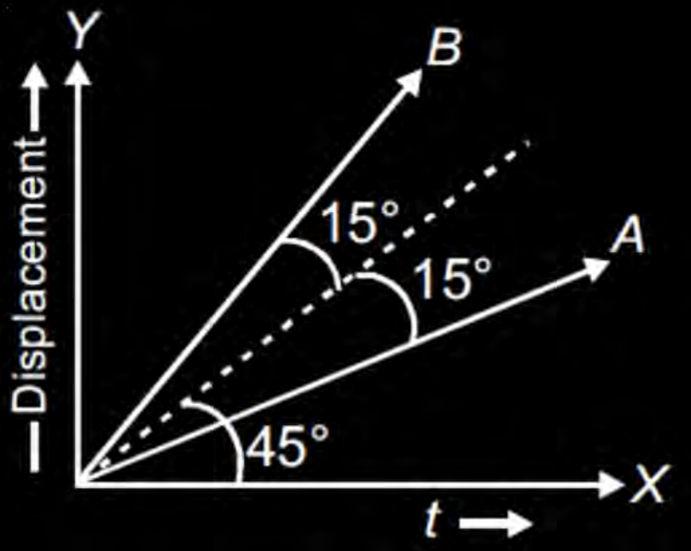


1. 10 m
2. 30 m
3. 13 m
4. 11 m

Question

The displacement-time graph for two particles A and B is follows. The ratio v_A/v_B is

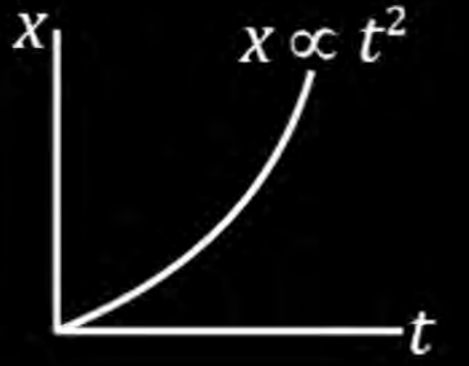
1. 1 : 2
2. $1 : \sqrt{3}$
3. 13 : 1
4. 1 : 3

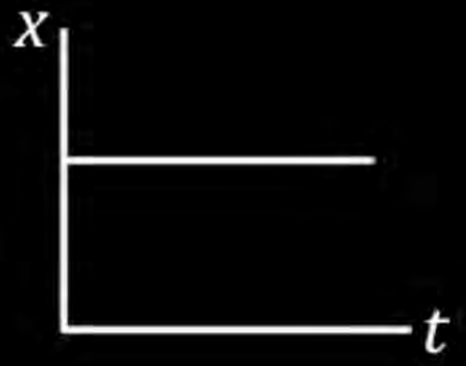


Question

In which graph acceleration is non-zero constant?

1. 

2. 

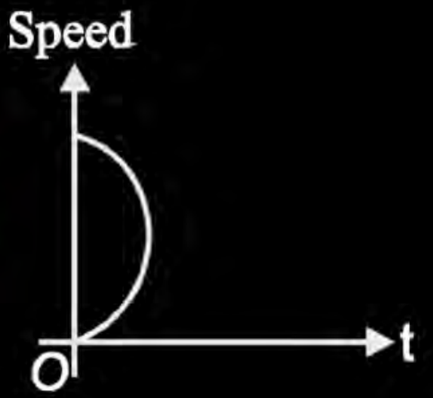
3. 

4. None of these

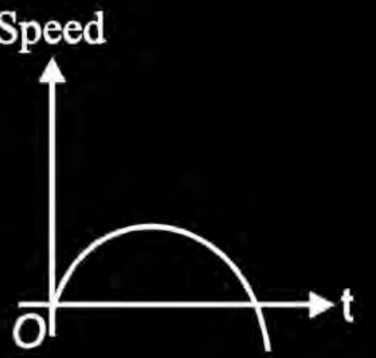
Question

Which one of the following graph for a body moving along a straight line is possible?

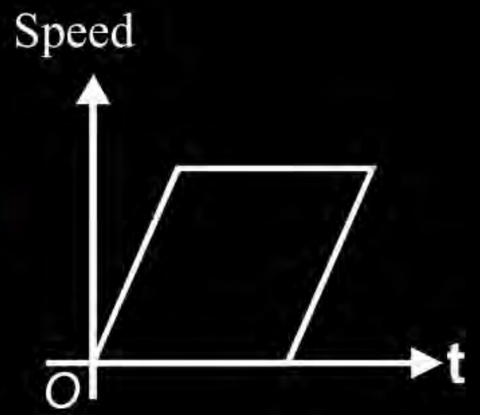
1



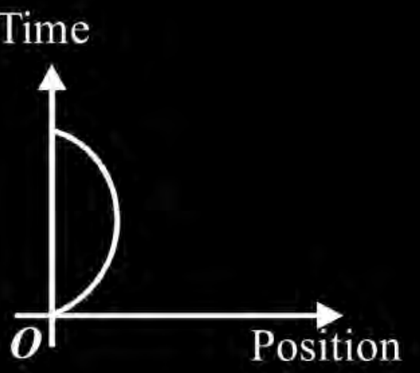
2



3

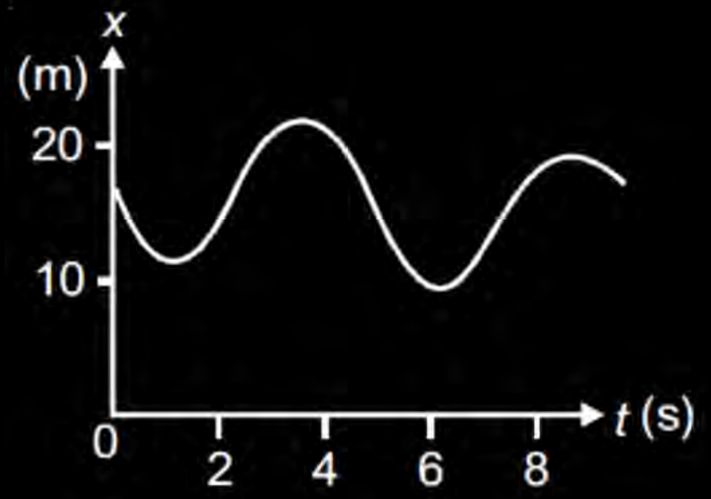


4



Question

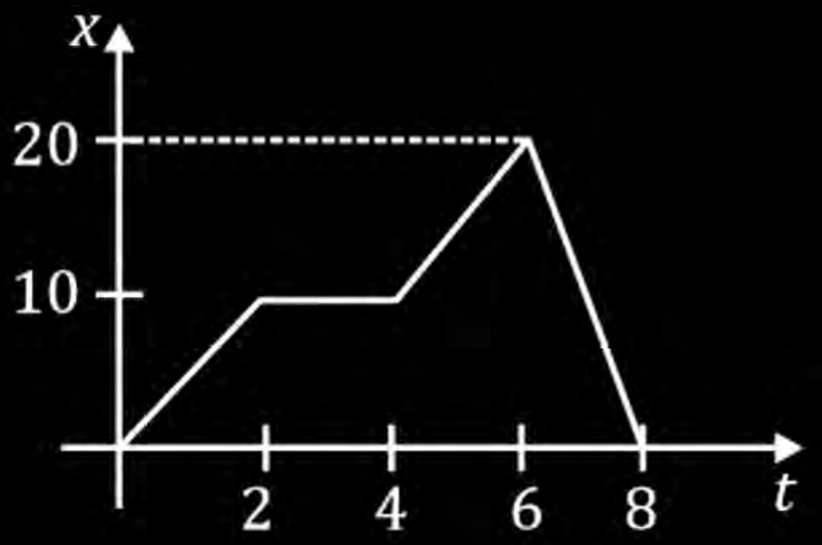
Figure shows the position of a particle moving on the x-axis as a function of time



1. The particle has come to rest 4 times
2. The velocity at $t = 8$ s is negative
3. The velocity remains positive for $t = 2$ s to $t = 6$ s
4. The particle moves with a constant velocity

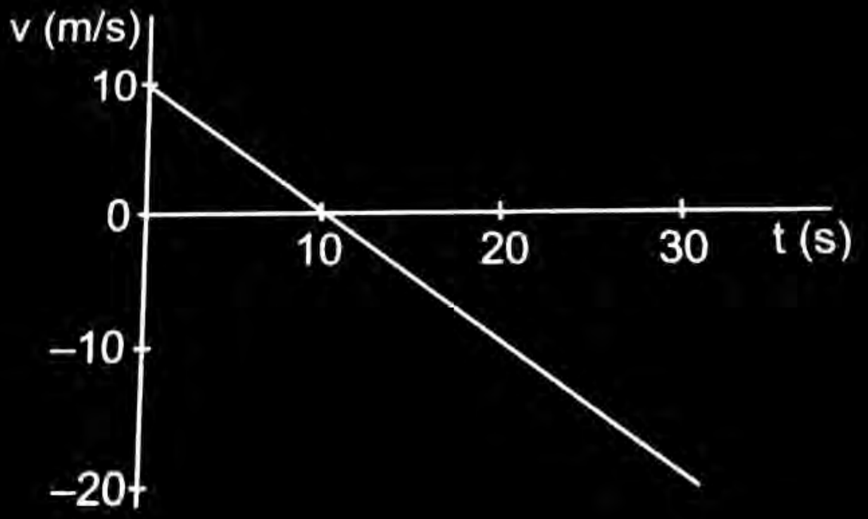
Question

Find avg speed and velocity in 8 sec.



Question

The velocity-time plot for a particle moving on a straight line shown in the figure



1. The particle has a constant acceleration
2. The particle has never turned around
3. The particle has a zero displacement
4. The average speed in the interval 0 to 10s is the same as the average speed in the interval 10s to 30s

Question

A car start from rest and moves with acceleration = 10 m/s^2 for time 5 sec after this it retards with –5 m/s^2 and comes to at rest then find total distance.

Question

Object moving with uniform velocity 20 m/s for 5 sec after that it retards with $-10\ m/s^2$ for 4 sec then find moves with constant for 2 more sec then find total distance and displacement.

Question

Object is moving with -10 m/s for 4 sec after that it accelerate $a = +5$ m/s^2 for 10 sec and then more for constant velocity for next 6 sec then find distance and displacement.

Question

A car starts from rest and moves with uniform acceleration a on a straight road from time $t = 0$ to $t = T$. After that, a constant deceleration brings it to rest. In this process the average speed of the car is

1. aT
2. $\frac{aT}{4}$
3. $\frac{3aT}{2}$
4. $\frac{aT}{2}$

Question

The engine of a motor cycle can produce maximum acceleration 5 m/s^2. It can produce maximum retardation 10 m/s^2 what is then minimum time in which he will cover a distance of 1.5 km.

1. 5 sec
2. 15 sec
3. 10 sec
4. 30 sec

Question

A car is moving with 20 m/s suddenly driver of car observes a stationary truck ahead of 100 m distance after some reaction time drive apply a break of conrt. retardation 4 m/s^2. find out maximum reaction time to prevent the collision?

PYQ on motion in a straight line

Solve only in rough Page

Maximum → 30 mint time duration

Question

The position x of a particle varies with time, (t) as $x = at^2 - bt^3$. The acceleration will be zero at time t is equal to: [1997]

1. $\frac{a}{3b}$
2. Zero
3. $\frac{2a}{3b}$
4. $\frac{a}{b}$

Question

The velocity of train increases uniformly from 20 km/h to 60 km/h in 4 hours. The distance travelled by the train during this period, is [1994]

1. 160 km
2. 180 km
3. 100 km
4. 120 km

Question

A car accelerates from rest at a constant rate α for some time after which it decelerates at a constant rate β and comes to rest. If total time elapsed is t, then maximum velocity acquired by car will be: **[1994]**

1. $\frac{(\alpha^2 - \beta^2)t}{\alpha\beta}$
2. $\frac{(\alpha^2 + \beta^2)t}{\alpha\beta}$
3. $\frac{(\alpha + \beta)t}{\alpha\beta}$
4. $\frac{\alpha\beta t}{\alpha + \beta}$

Question

A particle moves along a straight line such that its displacement at any time t is given by $s = (t^3 - 6t^2 + 3t + 4)$ metres. The velocity when the acceleration is zero is:

[1994]

1. 3 m/s
2. 42 m/s
3. –9 m/s
4. –15 m/s

Question

A body starts from rest, what is the ratio of the distance travelled by the body during the 4th and 3rd second? [1993]

1. 7/5
2. 5/7
3. 7/3
4. 3/7

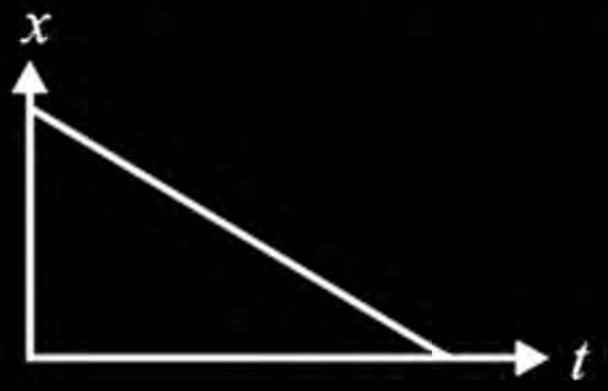
Question

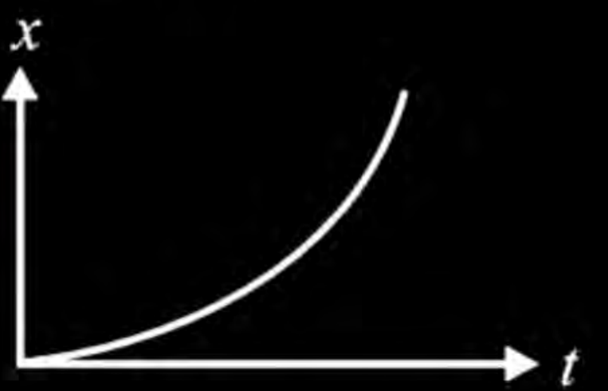
A car is moving along a straight road with a uniform acceleration. It passes through two points P and Q separated by a distance with velocity 30 km/h and 40 km/h respectively. The velocity of the car midway between P and Q is: **[MR* 1988]**

1. 33.3 km/h
2. $20\sqrt{2}$ km/h
3. $25\sqrt{2}$ km/h
4. 35 km/h

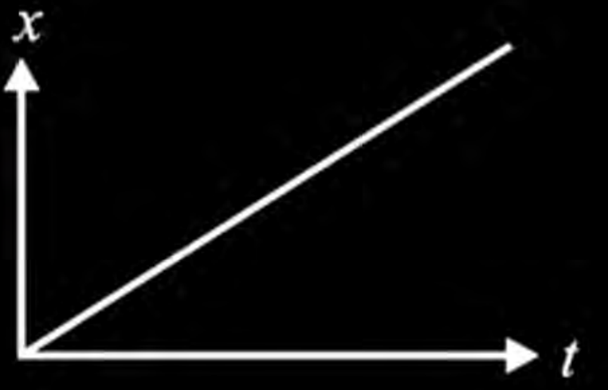
Question

The position-time (x-t) graph for positive acceleration is: [2022 Re]

1. 

2. 

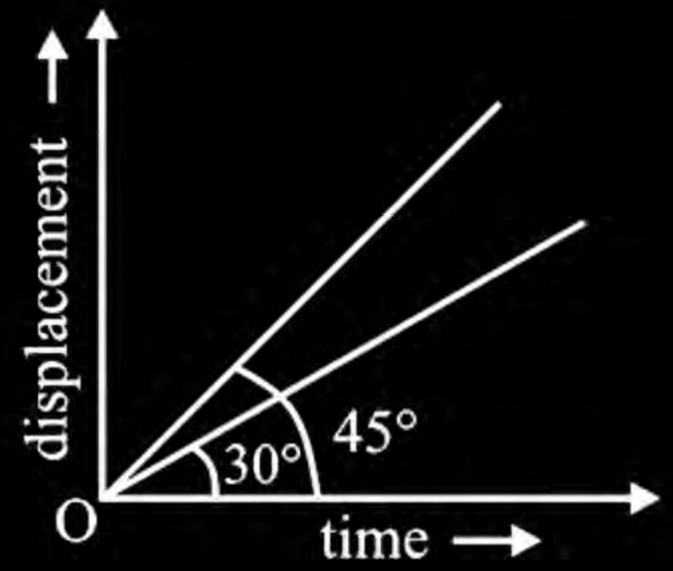
3. 

4. 

Question

The displacement time graphs of two moving particle make angles of 30° and 45° with the x-axis as shown in the figure. The ratio of their respective velocity is:

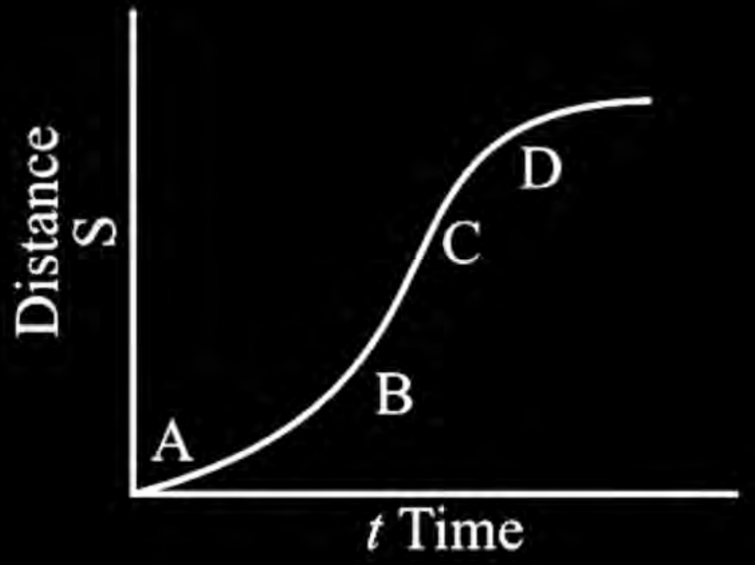
[2022]



1. $1 : \sqrt{3}$
2. $\sqrt{3} : 1$
3. $1 : 1$
4. $1 : 2$

Question

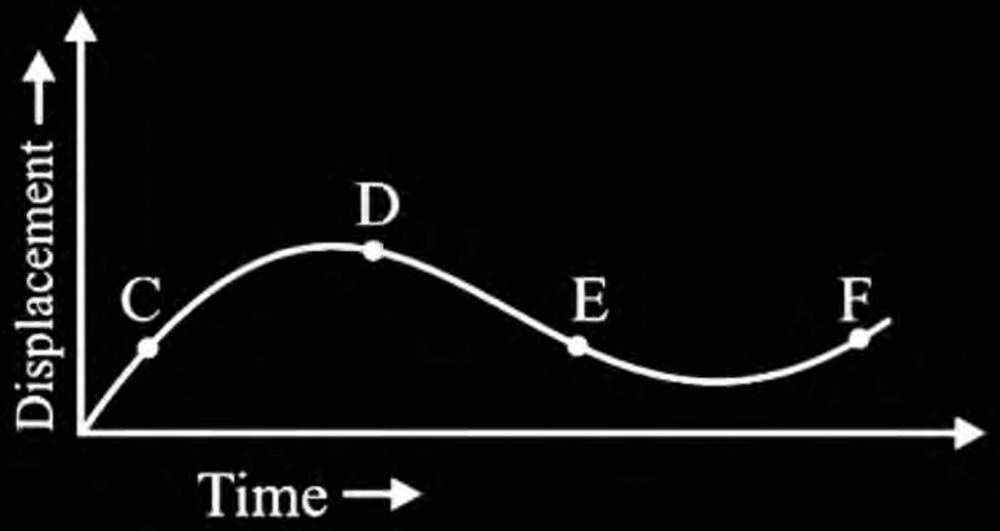
A particle shows distance-time curve as given in this figure. The maximum instantaneous velocity of the particle is around the point [2008]



1. D
2. A
3. B
4. C

Question

The displacement-time graph of a moving particle is shown below. The instantaneous of the particle is negative at the point: [MR* 1994]



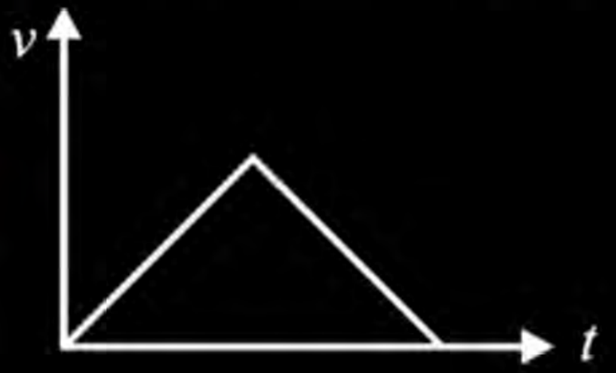
1. E
2. F
3. C
4. D

Question

Which of the following curve does not represent motion in one dimension?

[MR* 1992]

1



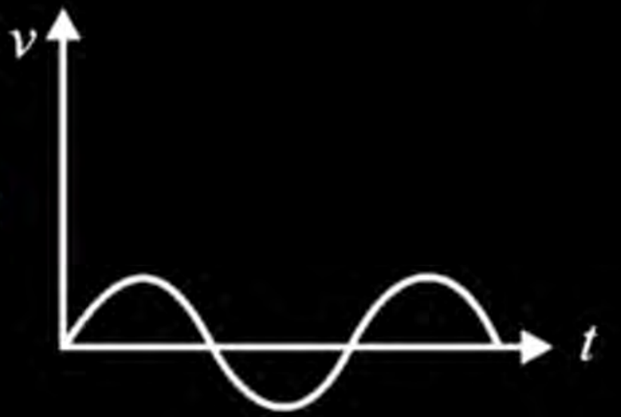
2



3



4



Question

The displacement x of a particle varies with time t as $x = ae^{-\alpha t} + be^{\beta t}$, where a, b, α and β are positive constants. The velocity of the particle will [2005]

1. be independent of β
2. drop to zero when $\alpha = \beta$
3. go on decreasing with time
4. go on increasing with time

Question

For a particle displacement time relation is $t = \sqrt{x} + 3$. Its displacement when its velocity is zero: [1999]

1. 2 m
2. 4 m
3. 0
4. None of these

Question

The acceleration of a particle is increasing linearly with time t as bt. The particle starts from origin with an initial velocity v_0. The distance travelled by the particle in time t will be: **[1995]**

1. $v_0 t + \frac{1}{3} b t^2$
2. $v_0 t + \frac{1}{2} b t^2$
3. $v_0 t + \frac{1}{6} b t^3$
4. $v_0 t + \frac{1}{3} b t^3$

Question

A bus travelling the first one-third distance at a speed of 10 km/h, the next one-third at 20 km/h and at last one-third at 60 km/h. The average speed of the bus is:

[MR* 1997]

1. 9 km/h
2. 16 km/h
3. 18 km/h
4. 48 km/h

Question

A car moves a distance of 200 m. It covers the first half of the distance at speed 40 km/h and the second half of distance at speed v. The average speed is 48 km/h. The value of v is: [1991]

1. 56 km/h
2. 60 km/h
3. 50 km/h
4. 48 km/h

Question

A car covers the first half of the distance between two places at 40 km/h and another half at 60 km/h. The average speed of the car is: [1990]

1. 40 km/h
2. 48 km/h
3. 50 km/h
4. 60 km/h

Question

A particle of unit mass undergoes one dimensional motion such that its velocity varies according to $v(x) = \beta x^{-2n}$ where β and n are constants and x is the position of the particle. The acceleration of the particle as a function of x, is given by: **[2015]**

1. $-2n\,\beta^2 x^{-4n-1}$
2. $-2n\,\beta^2 x^{+2n+1}$
3. $-2n\,\beta^2 x^{-4n+1}$
4. $-2n\,\beta^2 x^{-2n-1}$

Question

The motion of a particle along a straight line is described by equation: $x = 8 + 12t - t^3$ where x is in metre and t in second. The retardation of the particle when its velocity becomes zero, is **[2012 Pre]**

1. $24\ ms^{-2}$
2. Zero
3. $6\ ms^{-2}$
4. $12\ ms^{-2}$

Question

A particle moves a distance x in time t according to equation $x = (t + 5)^{-1}$. The acceleration of particle is proportional to **[2010 pre]**

1. $(\text{velocity})^{2/3}$
2. $(\text{velocity})^{3/2}$
3. $(\text{distance})^{2}$
4. $(\text{distance})^{-2}$

Question

A particle starts its motion from rest under the action of a constant force. If the distance covered in first 10 seconds is S_1 and that covered in the first 20 seconds is S_2, then: [2009]

1. $S_2 = 3S_1$
2. $S_2 = 4S_1$
3. $S_2 = S_1$
4. $S_2 = 2S_1$

Question

The distance travelled by a particle starting from rest and moving with an acceleration $\frac{4}{3}\,ms^{-2}$, in the third second is [2008]

1. $\frac{10}{3}\,m$
2. $\frac{19}{3}\,m$
3. 6 m
4. 4 m

Question

A particle moves in a straight line with a constant acceleration. It changes its velocity form 10 ms^{-1} to 20 ms^{-1} while passing through a distance 135 m in to second. The value of t is **[2008]**

1. 12
2. 9
3. 10
4. 1.8

Question

The position x of a particle with respect to time t along x-axis is given by $x = 9t^2 - t^3$ where x in metres and t in second. What will be the position of this particle when it achieves maximum speed along the +ve x direction? **[2007]**

1. 54 m
2. 81 m
3. 24 m
4. 32 m

Question

Motion of a particle is given by equation $S = (3t^3 + 7t^2 + 14t + 8)$m. The value of acceleration of the particle at $t = 1$ sec is: **[MR* 2000]**

1. $10\ m/s^2$
2. $32\ m/s^2$
3. $23\ m/s^2$
4. $16\ m/s^2$

Question

A particle starts from rest with constant acceleration. The ratio of average velocity to the time average velocity is: [2000]

1. 1/2
2. 3/4
3. 4/3
4. 3/2

Question

If a car at rest accelerates uniformly to a speed of 144 km/h in 20 sec, it covers a distance of [1997]

1. 1440 cm
2. 2980 cm
3. 20 m
4. 400 m

Question

A vehicle travels half the distance with speed v and the remaining distance with speed $2v$. Its average speed is: [2023]

1. $\frac{3v}{4}$
2. $\frac{v}{3}$
3. $\frac{2v}{3}$
4. $\frac{4v}{3}$

Question

Two cars P and Q start from a point at the same time in a straight line and their positions are represented by $X_P(t) = at + bt^2$ and $X_Q(t) = ft - t^2$. At what time do the cars have the same velocity? **[2016-II]**

1. $\frac{a+f}{2(1+b)}$
2. $\frac{f-a}{2(1+b)}$
3. $\frac{a-f}{1+b}$
4. $\frac{a+f}{2(b-1)}$

Question

If the velocity of a particle is $v = At + Bt^2$, where A and B are constants, then the distance travelled by it between 1s and 2s is: **[2016-I]**

1. $\frac{3}{2}A + 4B$
2. $3A + 7B$
3. $\frac{3}{2}A + \frac{7}{3}B$
4. $\frac{A}{2} + \frac{B}{3}$

Question

A particles covers half of its total distance with speed v_1 and the rest half distance with speed v_2. Its average speed during the complete journey is: **[2011 Mains]**

1. $\frac{v_1 v_2}{v_1 + v_2}$
2. $\frac{2v_1 v_2}{v_1 + v_2}$
3. $\frac{v_1^2 v_2^2}{v_1^2 + v_2^2}$
4. $\frac{v_1 + v_2}{2}$

Question

A car runs at a constant speed on a circular track of radius 100 m, taking 62.8 s for every circular lap. The average velocity and average speed for each circular lap respectively is: [2006]

1. 0, 0
2. 0, 10 m/s
3. 10 m/s, 20 m/s
4. 20 m/s, 0

Question

A particle moves along a straight line OX. At a time t (in seconds) the distance x (in meters) of the particle from O is given by $x = 40 + 12t - t^3$. How long would the particle travel before coming to rest? **[MR* 2006]**

1. 14 m
2. 16 m
3. 56 m
4. 40 m

Position of object $x = t^3 - 9t^2 + 16$ then find velocity when velocity is maximum.

Sol^n

Ans 27 m/s

Position of object $x = t^2 - 4t + 6$. then find distance and displacement in 3-sec.

dist^n = 5m
displacment = 3m

Object is moving with acceleration $a = 6 - 2v$ then find maximum velocity and velocity as a function of time.

gntial velocity was zero

Soln

Ans $v = 3(1 - e^{t})$

Object is moving such that Relation between time and Position $t = \alpha x^2 + \beta x$ then find acceleration if α and β are Constant.

(a) $2\alpha v^2$

(b) $-2\alpha\beta v$

(c) $-2\alpha v^3$

(d) $2\beta v^3$

If Position $x = (t+1)^{-1}$ then find acceleration as a function of time.

Ans $a \propto v^{3/2}$

Position of Particle is given as $x = 9t^2 - t^3$

find out distance travelled by Particle. when it achive maximum velocity...

54 m Ans

Position of Particle $x = a e^{-\alpha t} + b e^{\beta t}$

then find velocity with time

(a) $v \uparrow$

(b) $v \downarrow$

(c) $V = \text{Cost}^{n}$

(d) $a \downarrow$

(e) $a \uparrow$

gf velocity $v = \sqrt{108x - 9x^2}$ then find acceleration

Ans $a = (54 - 9x)$

if initial velocity of particle is u. find velocity as a function of time in acceleration given as follow.

(A) $a = kv$
find v

(B) $a = kv^2$
find v

(C) $a = kx^2$
find $u(x)$

Ans → $v = ue^{kt}$

Ans - $v = \frac{u}{1 - ukt}$

Ans - $u \propto x^{3/2}$

initial velocity of particle is $+V_0$ & retardation of particle is $a = -kx^3$ find displacement of particle when it comes to at rest.

Ans

$$x = \left(\frac{2V_0^2}{k}\right)^{1/4}$$

initial velocity of Particle is +6·25 m/s s retardation $\frac{dv}{dt} = -2.5\sqrt{v}$ find time when it comes to at rest.

Ans

$t = 2\,sec$

A Particle starts from rest at $t=0$. and acceleration $a = 4(2t-4)$ then find Velocity of Particle when acceleration is zero and velocity of Particle when velocity is maximum

Thank

You